Forecasting: Theory and Practice

Winter Semester 2026/27

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Overview

0. Organizational Matters

1. Introduction, Examples and Applications

2. Prediction Spaces, Calibration and Sharpness

References

0. Organizational Matters

Organization

Overview:

- *Specialization module* – 9 CP module
- Primary target audience: M.Sc. students of the department's programs (in particular Data Science, Mathematics, Business Mathematics)

What should you already know?

- Necessary: Introductory lecture to statistics (e.g., *Elementare Stochastik*)
- Recommended: *Machine Learning*, and some programming experience (ideally in `Python` or `R`, e.g., Internship Stochastics)
- Optional, but a plus: advanced statistics lecture(s) (e.g., *Measure and Integration Theory / Probability Theory, Statistics and Statistical Learning, Computational Statistics, ...*)

Lectures

The lecture takes place in person on

- Thursday, 16:15 - 17:45, HS VI (03A11)
- Friday, 10:15 - 11:45, HS VI (03A11)

Lecture will be recorded (but not live-streamed) and uploaded videos will be available on ILIAS, potentially for a limited period of time only.

Office hours: No fixed dates, please make an appointment by email, in which you briefly describe your case.

Exercises

There will be 6–7 bi-weekly exercise sheets, on which you can work as a team of up to 3 students.

The exercises will be a mix of theoretical and programming exercises.

We will implement a peer-feedback mechanism via ILIAS:

- Solutions are submitted as anonymized PDFs via ILIAS.
- In the following week, your team will be randomly assigned solutions by 2 other teams for providing feedback and comments.
- The feedback will then be visible in anonymized form to the team that submitted the solution.
- We will randomly check a subset of the submitted solutions and the submitted feedback.

Exercises and criteria for exam admission

Criteria for exam admission:

- Submission of solutions to all exercise sheets.
- Submission of feedback to all exercise sheets by other teams, which are assigned to your team.
- You are allowed one missed submission of either a solution to an exercise sheet or a feedback submission.
- If your submitted solution or feedback is subject to a random check and fails an assessment (e.g., in case of severely incorrect/incomplete solutions, or incorrect/inappropriate feedback and comments):
  - You will get one opportunity to revise your submitted solution / comments.
  - Your subsequent submissions will be checked regularly and carefully. Another failed check will result in failing to qualify for exam admission.

Exercises: Organizational details

Solutions and feedback have to be submitted via ILIAS until Wednesdays at 12:00 (only by one person designated by the team).

A tutorial (by Julius Günther) will take place on

- Wednesday, 14:15 - 15:45, SR II A3 (03A21)

starting on October 28, 2026.

Exemplary solutions to the exercises (for which feedback is to be submitted in the respective week) will be presented in the tutorial.

Exams

Exams: Oral or written exam, depending on the number of participants.

- Decision within the first few weeks of the semester
- exam registration via MARVIN
- Regulations, registration periods etc.: `https://www.uni-marburg.de/de/fb12/fachbereich/dekanat/pruefungsbuero/anmeldung-zu-modulpruefungen`

Prospective exam dates:

- First exam: 25.02.2027, HS B
- Second exam: 18.03.2027, HS B

Course material and updates

- all relevant material, information, news, and dates will be announced (only) via ILIAS, `https://ilias.uni-marburg.de/`
- Magazin > MARVIN: Kurse aller Semester > Wintersemester 2026/27 > FB12 Mathematik und Informatik > ...

Contact and communication

Contact details:

- Sebastian Lerch (lecture): sebastian.lerch@uni-marburg.de
- Julius Günther (exercises): julius.guenther@uni-marburg.de

There is a forum / discussion board which you can use

- for asking organizational questions of potential general interest
- to look for team members for the exercises,

Surveys

For interactive surveys and quizzes during the lectures, we will use PINGO, an open-source class room system, developed by the University of Paderborn.

To participate

- scan the QR code, or
- visit `https://pingo.coactum.de/` and enter access number 257256
- refresh the website to see new questions



Participation is anonymous and PINGO adheres to German/EU regulations for data processing and privacy.

Survey: Study programs

Which is your study program?

- BSc / MSc Data Science
- BSc / MSc Computer Science
- BSc / MSc Mathematics
- BSc / MSc Business Informatics
- Lehramt
- Other

Survey: Prerequisites

Which other lectures did you attend already? (multiple choice)

- Elementary Stochastics or other introduction to statistics
- Statistics and Statistical Learning
- Computational Statistics
- Other advanced lecture on probability theory or statistics
- Internship Stochastics
- Machine Learning
- Advanced lecture on Machine Learning

Survey: Programming experience

How much experience do you have with R?

- No programming experience at all
- No experience, but have used other programming languages
- Basic knowledge, but need to look up most basic functions
- More advanced knowledge

Literature

- These slides are partly based on (unpublished) lecture notes by Tilmann Gneiting (Karlsruhe Institute of Technology).
- References to original literature are provided throughout and serve as primary sources.
- Overview articles:
  - Gneiting and Katzfuss (2014)
  - Petropoulos et al. (2022)
  - Klein (2024)

1. Introduction, Examples and Applications

Overview

1. Introduction, Examples and Applications

1.1 Probabilistic and point forecasts

1.2 Examples and applications

1.3 Forecasting competitions and prediction markets

Forecasting

"It's difficult to make predictions, especially about the future."

Danish proverb, frequently attributed to Niels Bohr.

A major human desire is to make forecasts for an uncertain future.

Consequently, forecasts ought to be probabilistic in nature, taking the form of probability distributions over future quantities or events.

Over the past decade(s), a transition from point prediction to distributional prediction, or probabilistic forecasting, can be observed across applications.

Forecasting

- Suppose we are trying to predict $Y \in \mathcal{Y}$
  - Binary outcomes: $Y \in \{0, 1\}$
  - Categorical outcomes: $Y \in \{1, 2, \ldots, m\}$
  - Univariate outcomes: $Y \in \mathbb{R}$
  - Multivariate outcomes: $Y \in \mathbb{R}^d$
- Predictions come in many forms
  - A point prediction is a single value $x \in \mathcal{Y}$
  - A probabilistic prediction is a probability distribution F over $\mathcal{Y}$
  - A prediction interval (or prediction set) is a set $I \subset \mathcal{Y}$
- We will focus mainly on probabilistic predictions
  - Point predictions can be derived from probabilistic predictions
  - e.g., by taking the mean, median, or a quantile
  - Prediction intervals can also be derived from probabilistic predictions

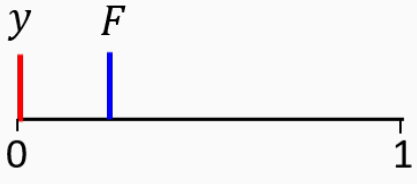
Forecasting

- Example: Binary outcomes



(a) Point forecast

$x \in \{0, 1\}$

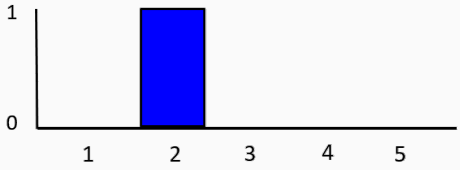


(b) Probabilistic forecast

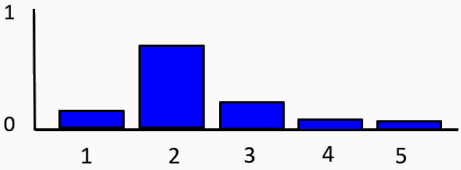
$F \in [0, 1]$

Forecasting

- Example: Categorical outcomes



(a) Point forecast

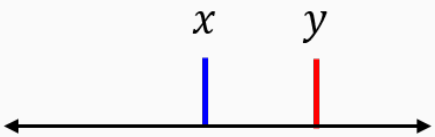


(b) Probabilistic forecast

$x \in \{1, 2, \ldots, m\}$

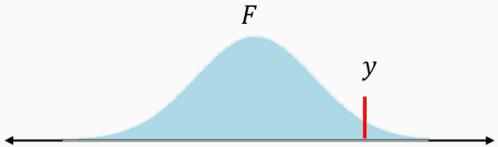
$F \in [0,1]^m \quad \text{with} \quad \sum_{j=1}^{m} F_j = 1$

- Example: Univariate outcomes



(a) Point forecast

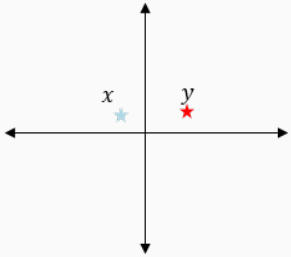
$x \in \mathbb{R}$



(b) Probabilistic forecast

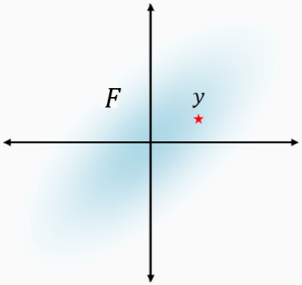
F is a probability distribution on $\mathbb{R}$

- Example: Multivariate outcomes



(a) Point forecast

$x \in \mathbb{R}^d$



(b) Probabilistic forecast

F is a probability distribution on $\mathbb{R}^d$

Probabilistic vs. point predictions

We make predictions to obtain useful information about the future that we can use to aid decision making and risk assessment.

Probabilistic predictions quantify our uncertainty and are more useful for decision making.

Point forecasts are typically easier to communicate, and most applications require "deterministic" decisions:

- cancel an event or not
- build a dam with a specific height
- sell or hold a position in an investment portfolio
- operate a water reservoir
- ...

Learning goals

The overarching aim of the lecture is to learn how to make and evaluate probabilistic forecasts.

We will cover

- how probabilistic forecasts are used across applications, from economics to weather and climate
- mathematical foundations of how probabilistic (and point) forecasts can be evaluated
- how we can design, implement and estimate statistical and machine learning methods that yield probabilistic forecasts

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Macroeconomic forecasting

Process of predicting economy-wide performance and major indicators like gross domestic product (GDP), inflation, and unemployment.

Forecasts are often made in quarter-yearly resolution, with time horizons short-term updates to multi-year projections.

Common methods for producing such forecasts (for an overview, see Stock and Watson, 2006) include

- theory-driven structural models
- data-driven statistical models, e.g., time series models
- more recently also machine learning models

We will consider two example, the inflation forecasts by the Bank of England, and the Survey of Professional Forecasters by the European Central Bank.

Bank of England inflation forecasts

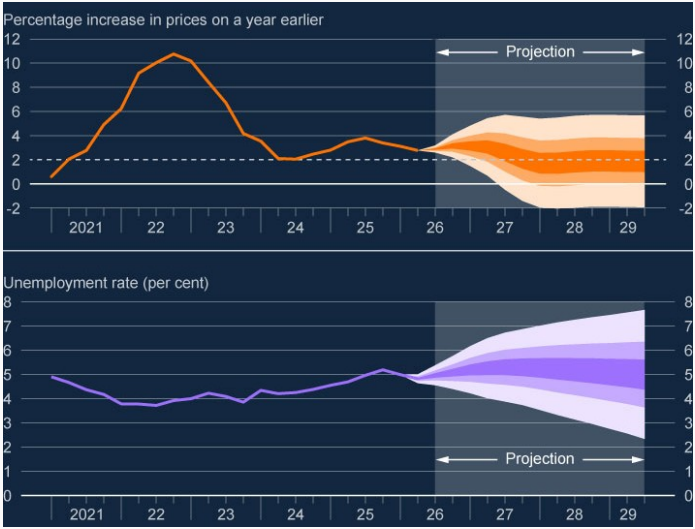
The Bank of England's Monetary Policy Committee (MPC) has issued probabilistic forecasts of inflation rates every quarter since February 1996.

Fan charts are used to visualize selected deciles of the predictive distributions.

The fan charts are constructed using historical forecast errors as the empirical basis, with the MPC judgementally adjusting the width and skew of the distribution.

Current forecasts are available at
`https://www.bankofengland.co.uk/monetary-policy-report/2026/july-2026`.

Bank of England inflation forecasts



https://www.bankofengland.co.uk/monetary-policy-report/2026/july-2026, Section 3.2

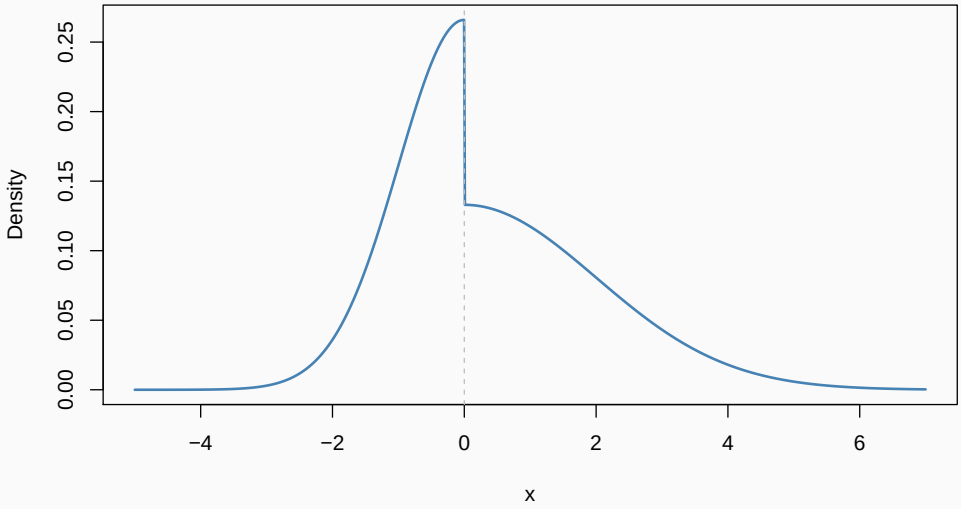
Bank of England inflation forecasts

The MPC employs potentially asymmetric two-piece normal distributions with parameters $\mu \in \mathbb{R}$ and $\sigma_1, \sigma_2 > 0$ and forecast density

$$f(y) = \begin{cases} \left(\frac{\pi}{2}\right)^{-1/2} (\sigma_1 + \sigma_2)^{-1} \exp\left(-\frac{(y-\mu)^2}{2\sigma_1^2}\right) & \text{if} \quad y \leq \mu, \\ \left(\frac{\pi}{2}\right)^{-1/2} (\sigma_1 + \sigma_2)^{-1} \exp\left(-\frac{(y-\mu)^2}{2\sigma_2^2}\right) & \text{if} \quad y \geq \mu. \end{cases} \tag{1.1}$$

In the exercises, you will find the expectation and variance of this density and prove that it is right-skewed when $\sigma_2 > \sigma_1$.

Bank of England inflation forecasts



European Central Bank Survey of Professional Forecasters

There is a long history of systematic surveys of professional economists' macroeconomic forecasts, in particular in the US.

The Survey of Professional Forecasters (SPF) is a quarterly survey conducted by the European Central Bank among professional forecasters, which has been running since 1999.

- conducted in January, April, July and October
- respondents are economists/forecasters working at financial institutions, research institutions and other organizations
- main variables (probabilistic forecasts): inflation, GDP growth, unemployment rate
- secondary variables (point forecasts): wages, USD/EUR exchange rate, oil prices, and others

More details: `https://www.ecb.europa.eu/press/economic-bulletin/html/eb201901~a3afea780e.en.html#IDofArticle1`

European Central Bank Survey of Professional Forecasters

For the main variables, respondents provide a probability distribution over possible outcomes.

An individual response is a point forecast plus a histogram-like subjective predictive distribution, where the bins have a width of 0.5 percentage points, e.g.,

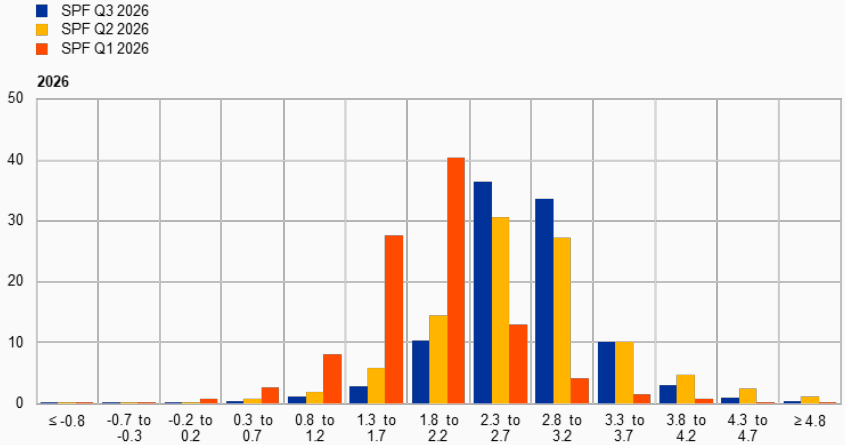
Inflation rate	< 0	$0-0.5$	$0.5-1.0$	$1.0-1.5$	$1.5-2.0$	$2.0-2.5$	$2.5-3.0$	$3.0-3.5$	≥ 3.5
$\mathbb{P}_i(Y \in B_k)$	0.01	0.02	0.04	0.10	0.25	0.32	0.16	0.07	0.03

The ECB aggregates these individual probability assessments into aggregate probability distributions.

Current forecasts are available at `https://www.ecb.europa.eu/stats/ecb_surveys/survey_of_professional_forecasters/html/ecb.spf2026q3.en.html`.

European Central Bank Survey of Professional Forecasters

Aggregate probability distributions for expected inflation in 2026:



https://www.ecb.europa.eu/stats/ecb_surveys/survey_of_professional_forecasters/html/ecb.spf2026q3.en.html

Other economic forecasting applications

Point and probabilistic forecasts are highly relevant in many areas of economics and business, for example

- exchange rate forecasting
- financial forecasting (e.g., stock prices, portfolio management)
- risk management
- supply chain management (demand and production forecasting)
- forecasting new product success
- predictive maintenance

Overview and more applications: Petropoulos et al. (2022).

Example: Risk management

The expected shortfall or α-conditional value-at-risk functional (CVaR_α, $0 < \alpha < 1$) equals the expectation of a random variable X conditional on it taking values in its upper $(1-\alpha)$-tail, i.e.,

$$\mathrm{CVaR}_\alpha(X) = \frac{1}{1-\alpha}\mathbb{E}\left[X \cdot \mathbf{1}_{X \geq q_\alpha(X)}\right],$$

where q_α denotes the α-quantile.

The CVaR a popular risk measure in quantitative finance and plays a key role in the Basel III - regulations of the Basel Committee on Banking Supervision of the Bank for International Settlements (BIS) governing the regulation of banks.

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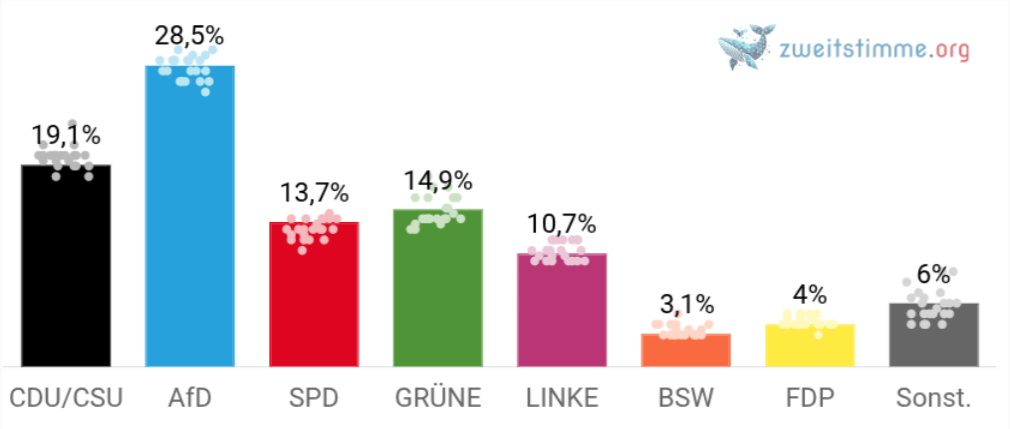
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Election forecasting

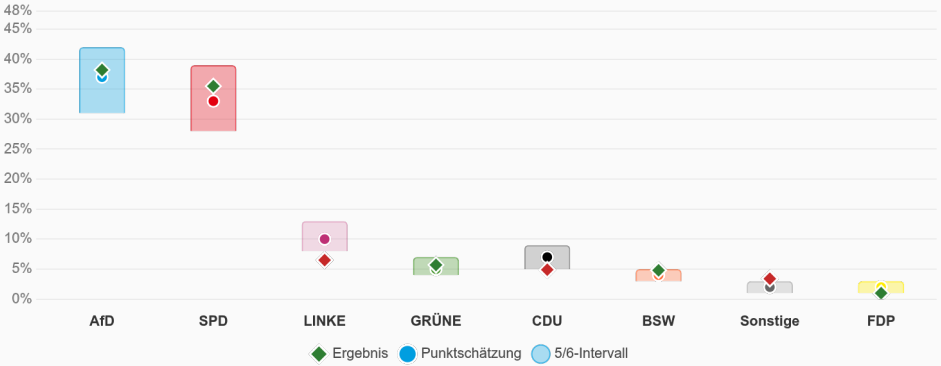
Opinion polls are widespread, but can/should not be interpreted as forecasts (and usually offer no probabilistic information).



https://zweitstimme.org/, accessed October 5, 2026

Election forecasting

Some platforms (e.g., FiveThirtyEight (now discontinued), Zweitstimme.org) produce probablistic election forecasts:



https://zweitstimme.org/blog/posts/be-mv-2026-evaluation/,

Probabilistic forecasts for the state parliament election in Mecklenburg-Vorpommern in September 2026

Challenges in political forecasting

- small sample sizes, with, e.g., national elections occurring every 4-5 years only
- data scarcity
- complex, non-linear reactions to events (e.g., economic shocks)
- feedback loops: forecasts may influence the outcomes, e.g., due to
  - voter behavior (e.g., strategic voting)
  - turn-out dynamics
  - donor behavior

Conflict early warning and violent unrest

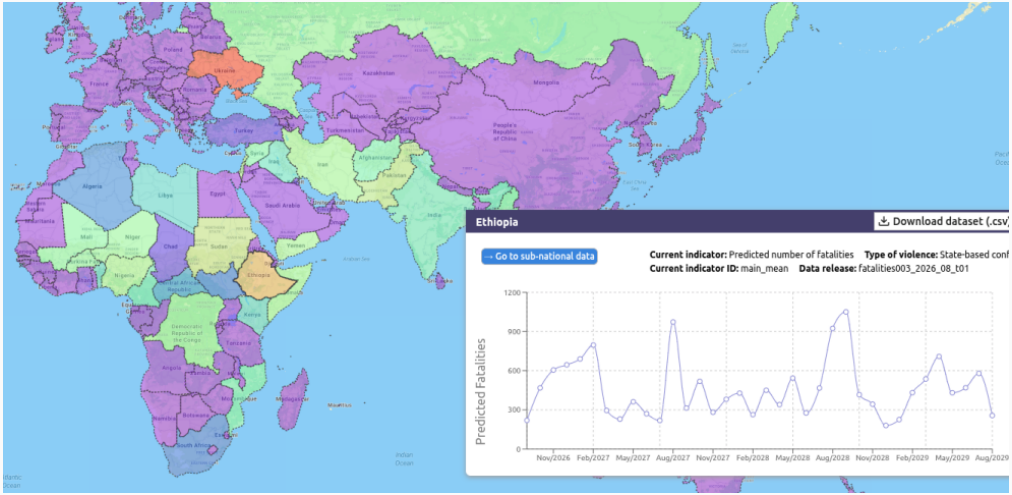
Governments, international agencies, and NGOs utilize (probabilistic) forecasts to allocate humanitarian aid and diplomatic resources.

Often event-based forecasting, i.e., predicting the probability of conflict onset, escalation, or termination in specific grid cells or countries over a 1–12 month horizon.

ViEWS (Violence & Impacts Early-Warning System) Consortium:

- runs "prediction challenges" and produces deterministic and probabilistic forecasts, e.g., of the number of fatilities in armed conflict
- platform collecting forecast and observation data
- details: Hegre et al. (2019)

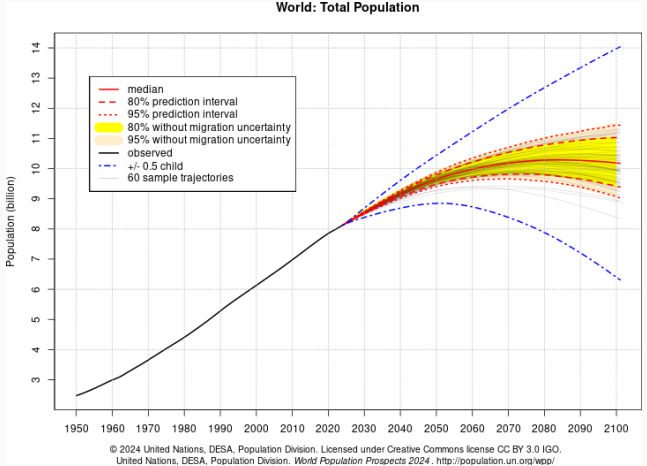
Conflict early warning and violent unrest



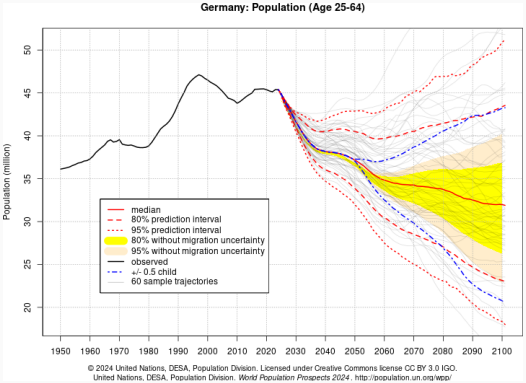
https://data.viewsforecasting.org/

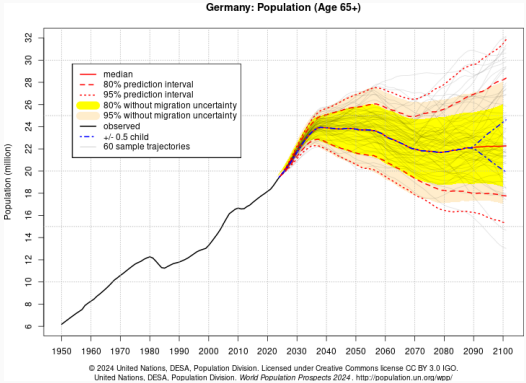
Demography

Probabilistic population predictions are available from the United Nations. Details on methodology: Raftery et al. (2014)



UN population predictions for Germany





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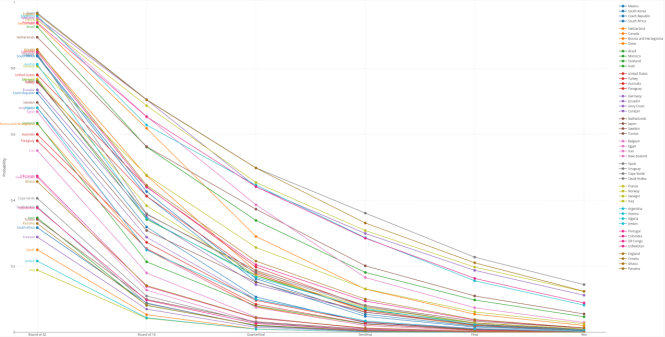
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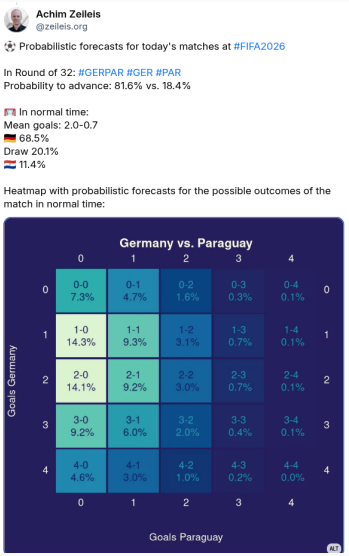
Match- and tournament-level prediction

Probabilistic modeling of match-level outcomes is of interest to betting markets, team management, and fan media.

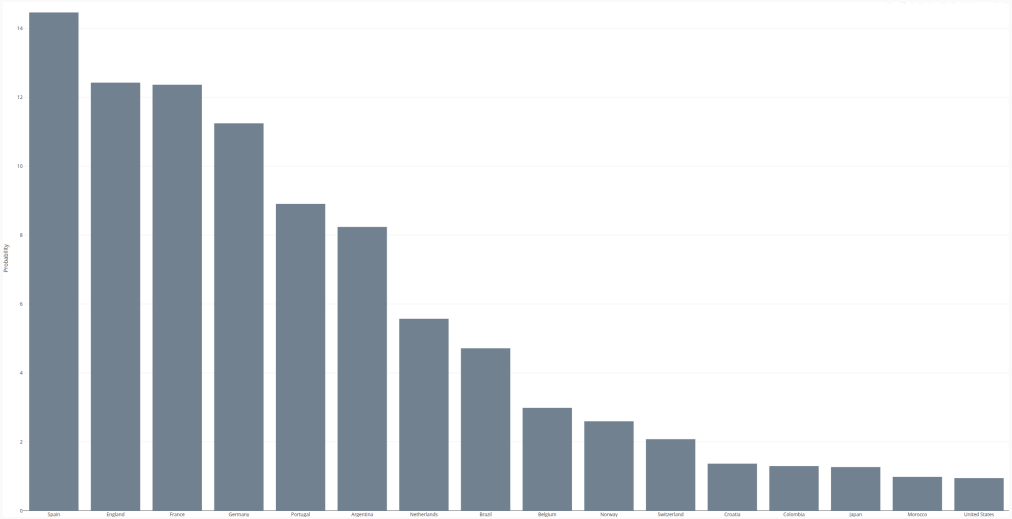
Simulated games based on estimated winning probabilities enable tournament-level predictions.



https://www.zeileis.org/news/fifa2026/#winning-probabilities



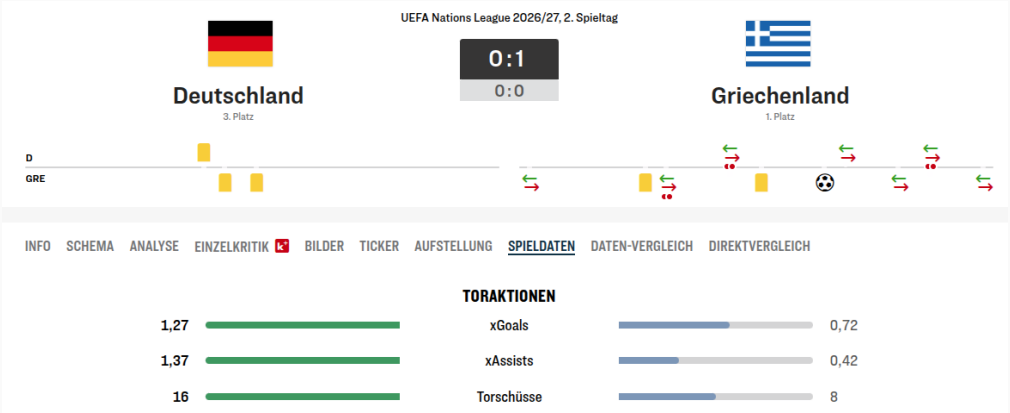
Match- and tournament-level prediction



https://www.zeileis.org/news/fifa2026/#winning-probabilities

Event-level probabilities

Tracking data allows to evaluate individual situations, e.g., estimates of the probability of a goal given the shot location, defender positions, etc. to compute a team's expected goals



https://www.kicker.de/griechenland-gegen-deutschland-2026-uefa-nations-league-5193276/spieldaten

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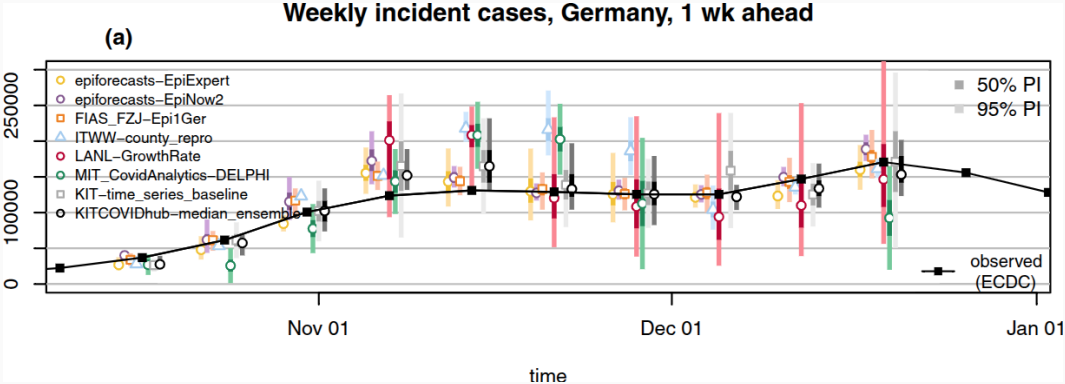
Health forecasting

Important health domain applications include

- epidemiology:
  - targets: incident cases, reproduction number (R_t), deaths
  - usually on the level of regions or entire countries
  - highly relevant during the COVID-19 pandemic for political response and management
- operational healthcare forecasting at hospitals, intensive care units (ICUs), etc.
  - targets: occupancy, length of stay (LoS), arrivals
  - patient-level risks and outcomes
  - increasingly informed by epidemiological modeling (e.g. during the COVID-19 pandemic)

Example: COVID-19 forecasting hubs

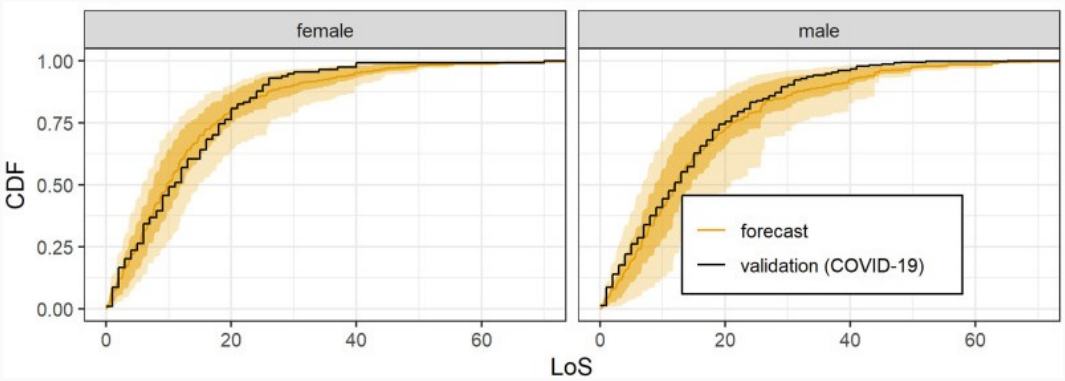
Starting with the US, forecasting hubs became available as open comparison platform for teams to submit probabilistic, real-time model predictions.



Details: Bracher et al. (2021)

Example: Intensive care unit (ICU) length of stay (LoS)

Probabilistic LoS prediction of patients in Swiss ICUs with the aim of improving resource allocation and patient care during the COVID-19 pandemic:



Details: Henzi et al. (2021)

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1.2.5 Energy forecasting

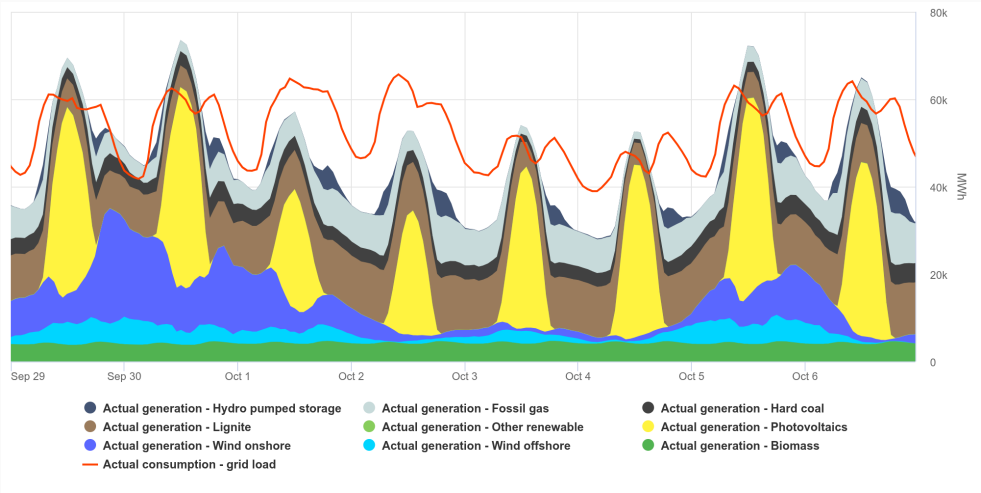
1.2.6 Weather and climate

1.2.7 Other environmental sciences applications

1.3 Forecasting competitions and prediction markets

Energy generation in Germany

Energy generation in Germany is highly weather-dependent, with high variability



https://www.smard.de/en/marktdaten/

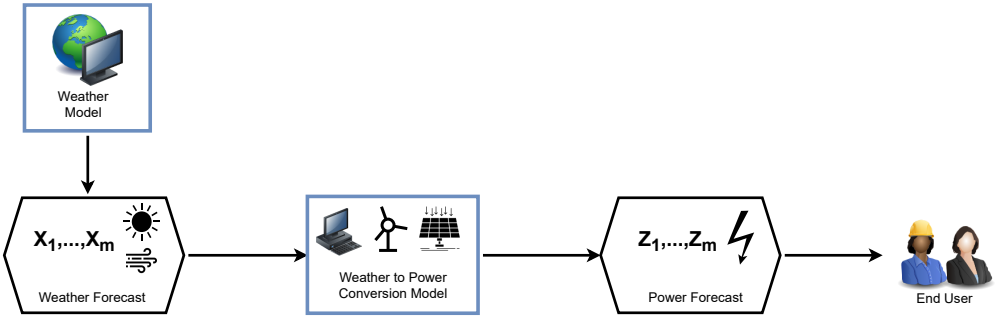
Energy forecasting

Energy forecasting is relevant for several applications, including

- renewable energy generation (in particular wind and solar energy)
- electric load (i.e., consumption)
- electricity prices
- power system planning (e.g., battery storage) and operation (e.g., maintenance)

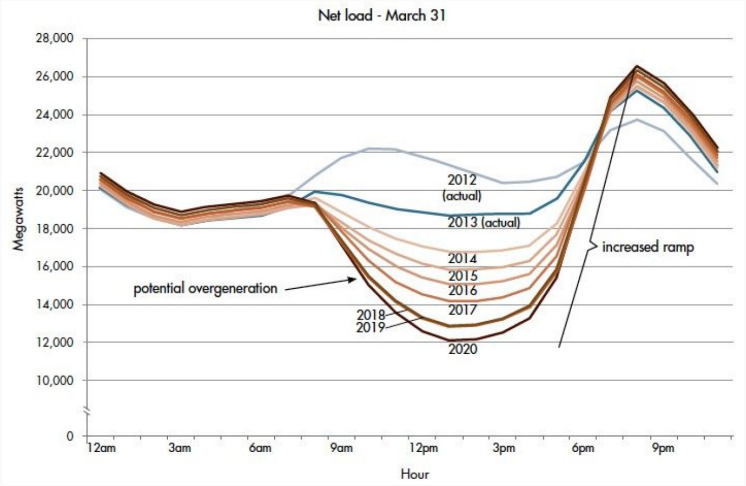
Overview: Hong et al. (2020)

Renewable energy generation forecasting



Electric load forecasting

"Duck curve effect": Net load (i.e., difference of actual load and solar energy generation) in California:



https://www.energy.gov/cmei/solar/

Electricity price forecasting

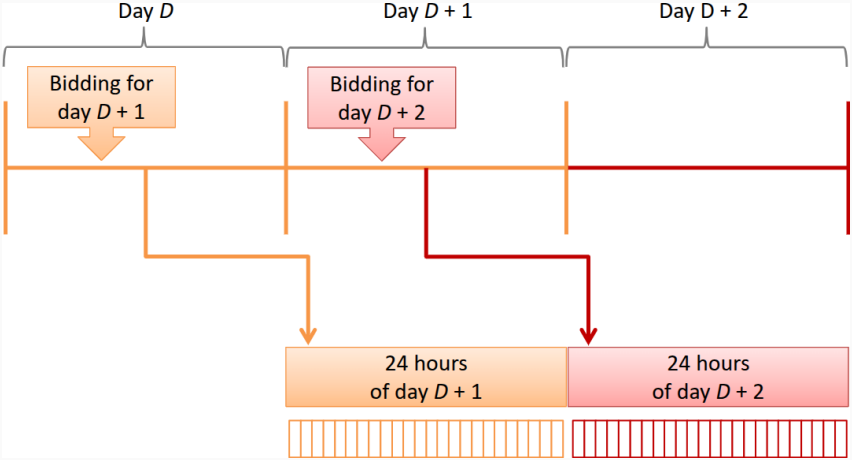
Electricity can be traded on various wholesale spot markets in Europe



https://www.epexspot.com/en/basicspowermarket

Day-ahead electricity price forecasting

The day-ahead market is of particular importance



Electricity price forecasting (EPF)

EPF is important because it cannot be stored at scale easily or cheaply, and supply and demand must match continuously in real time across an interconnected physical grid.

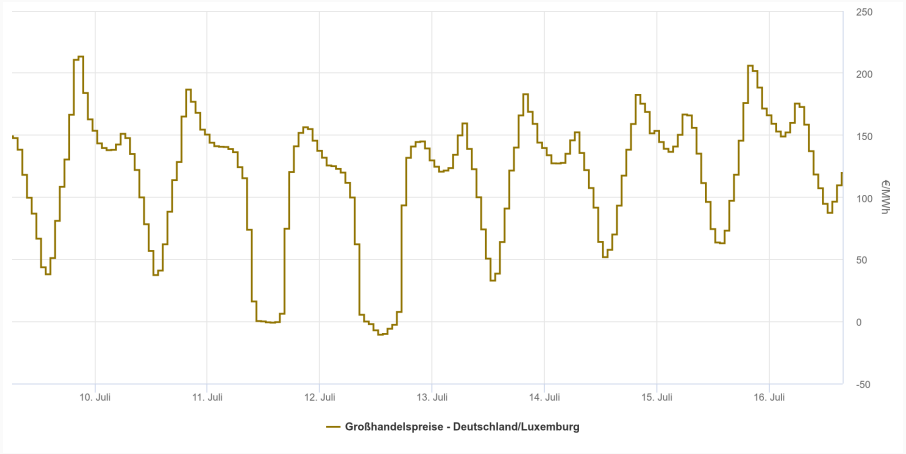
Probabilistic forecasts are particularly important for risk management.

Market participants include

- Generators of electricity, who need to decide whether to clear their marginal generation costs or withhold capacity, e.g. for scheduling maintenance or avoiding negative prices
- Retailers & utilities, who need to build competitive fixed- or dynamic-rate retail tariffs for consumers while covering market risks
- Proprietary traders

Electricity price forecasting (EPF)

As renewable energy generation increases, hours of oversupply regularly lead to negative electricity prices (where generators pay off-takers to consume power).



https://www.smard.de/home/marktdaten/

Electricity price forecasting (EPF)

Challenges in intra-day and day-ahead EPF include

- sudden, severe jump dynamics (spikes) in times when demand exceeds cheap generation, and high-cost peaking units (e.g., gas turbines) are triggered
- negative prices
- overlapping, complex seasonal patterns (intra-day, weekly, annual)
- structural breaks (regulatory interventions, fuel price shifts, etc.)
- dependence on exogenously uncertain features (weather forecasts)

Overviews of challenges, forecasting methods, and applications: Weron (2014); Nowotarski and Weron (2018)

Overview

1. Introduction, Examples and Applications

1.1 Probabilistic and point forecasts

1.2 Examples and applications

1.2.1 Economic forecasting

1.2.2 Political and social science

1.2.3 Sports analytics

1.2.4 Health and epidemiology

1.2.5 Energy forecasting

1.2.6 Weather and climate

1.2.7 Other environmental sciences applications

1.3 Forecasting competitions and prediction markets

Weather forecasts and their societal and economic importance

Weather forecasts have substantial economic and societal value:

- Public safety: early warnings to enable preparation for storms, floods, heatwaves and other hazardous events, saving lives and reducing damage.
- Economic decisions for weather-dependent industries such as agriculture, energy production and trading, transportation, construction, or insurance
- Operational planning to help allocate resources, schedule activities and manage weather-related risks
- Everyday life and personal decisions

Weather: Forecast data

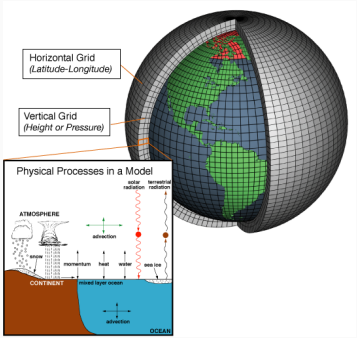


Figure: NOAA

Modern weather forecasts rely on physical models of the atmosphere.

Development of the state of the atmosphere on a 3D grid:

time t
initial conditions



physical laws
$\longrightarrow$
partial differential equations

time $t+1$
prediction



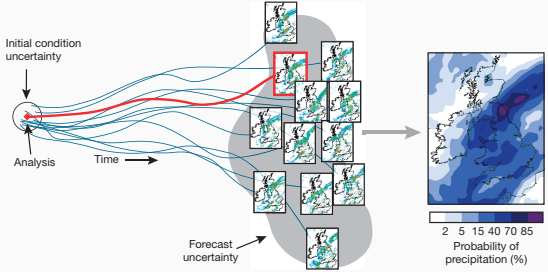
physical laws
$\longrightarrow$
partial differential equations

time $t+2$
prediction



Ensemble forecasts

Ensemble simulations allow for quantifying forecast uncertainties due to initial conditions, model errors, and non-linear error growth in a chaotic system.



Bauer et al. (2015)

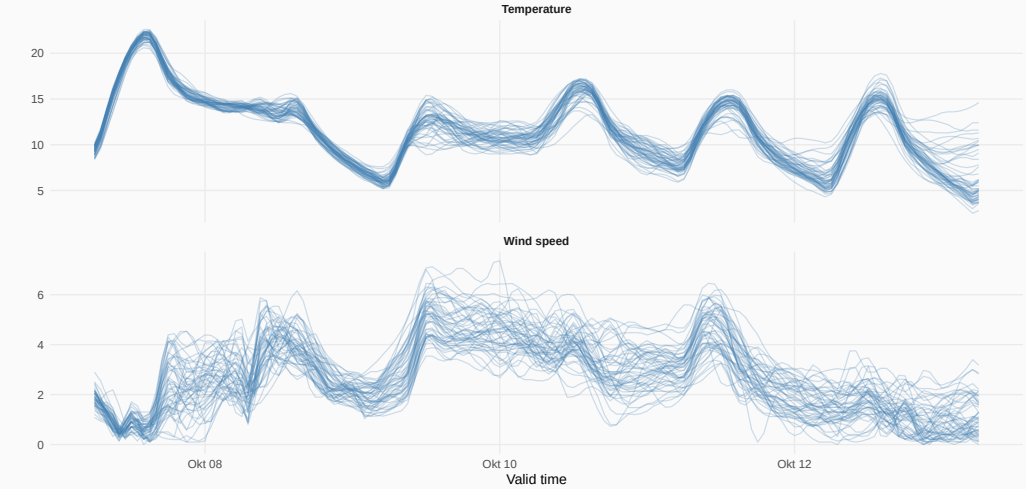


ECMWF

Overview, history, and further reading: Bauer et al. (2015)

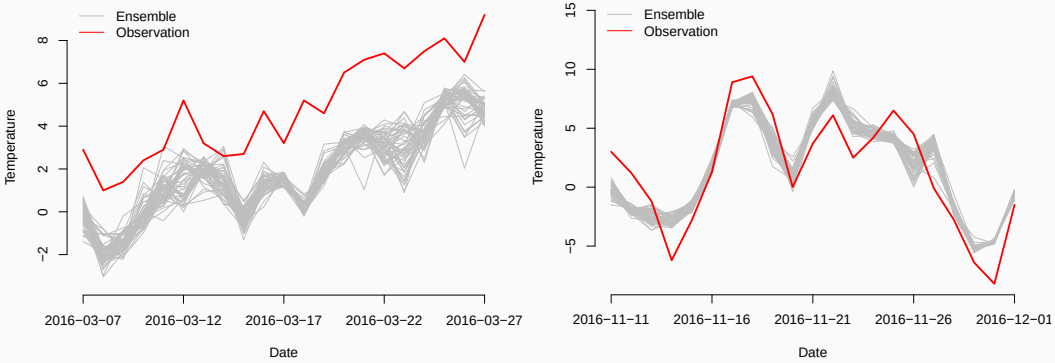
Ensemble forecasts

Exemplary ensemble forecasts for Frankfurt airport, initialized on October 7, 2026 at 06:00 (UTC)



Typical errors of ensemble weather forecasts

48-hour ahead ECMWF ensemble forecasts of 2-meter temperature at Frankfurt Airport, Germany



Post-processing ensemble weather forecasts

As ensemble forecasts are subject to biases and dispersion errors, some form of statistical post-processing is required.

Given ensemble forecasts of 2-meter temperature,

$$\boldsymbol{X}^{\text{t2m}} = (x_1, ..., x_M),$$

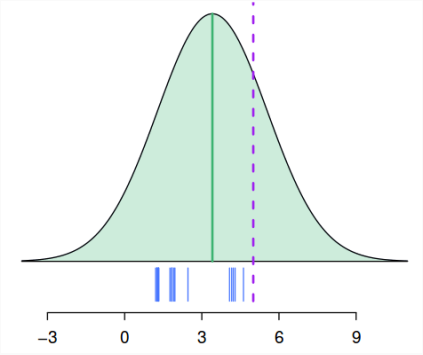
(parametric) post-processing approaches model the conditional distribution of the target variable y as a parametric distribution,

$$y \mid x_1, ..., x_M \sim F_\theta,$$

with the aim of correcting systematic errors by exploiting structure in past forecast-observation pairs.

Post-processing ensemble weather forecasts: EMOS

One popular approach is ensemble model output statistics (EMOS; Gneiting et al., 2005), where F_θ is a single parametric distribution. For example, for temperature forecasting the post-processed forecast takes the form of a Gaussian distribution.



$$y|\boldsymbol{X}^{\text{t2m}} \sim \mathcal{N}_{(\mu,\sigma)},$$

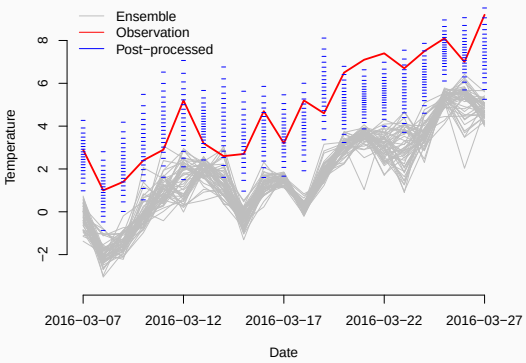
$$\mu = a + b \cdot \text{mean}(\boldsymbol{X}^{\text{t2m}})$$

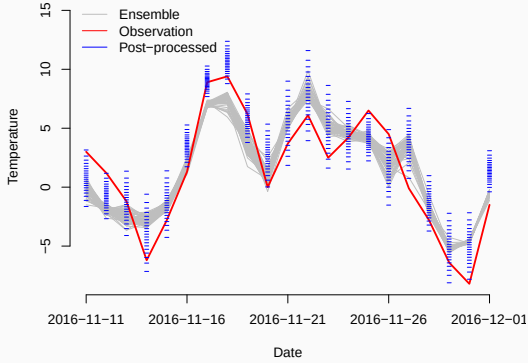
$$\sigma = c + d \cdot \text{sd}(\boldsymbol{X}^{\text{t2m}})$$

The EMOS coefficients a, b, c, d need to be estimated from training data.

Post-processing ensemble weather forecasts: EMOS

48-hour ahead ECMWF ensemble forecasts of 2-meter temperature at Frankfurt Airport, Germany





Post-processing ensemble weather forecasts: EMOS

If the ensemble forecasts $x_1, ..., x_M$ can not be considered exchangeable (e.g., if they are produced by different weather models), an alternative model formulation is

$$y \,|\, x_1, \ldots, x_M \sim \mathcal{N}(a_0 + a_1 x_1 + \cdots + a_M x_M,\ b_0 + b_1 s),$$

with bias parameters a_0, $a_1, \ldots, a_M$, and spread parameters b_0 and b_1, where s is the standard deviation of the ensemble values.

Post-processing ensemble weather forecasts: EMOS

EMOS specifications for continuous weather quantities:

Weather quantity	Range	Predictive distribution(s)
Temperature	$y \in \mathbb{R}$	Normal; skew-normal; mixtures of normals
Pressure	$y \in \mathbb{R}$	Normal
Precipitation amount	$y \in \mathbb{R}_+$	Left-censored GEV; censored, shifted gamma
	$y^{1/2} \in \mathbb{R}_+$	Left-truncated logistic
Wind components (u, v)	$(u, v) \in \mathbb{R}^2$	Bivariate normal
Wind speed	$y \in \mathbb{R}_+$	Left-truncated normal; left-truncated logistic; log-normal; GEV
Wind gusts	$y \in \mathbb{R}_+$	Left-truncated normal; left-truncated logistic; GEV

Post-processing ensemble weather forecasts: BMA

Bayesian model averaging (BMA; Raftery et al., 2005) employs a mixture distribution, where each component is a parametric density associated with an individual ensemble member.

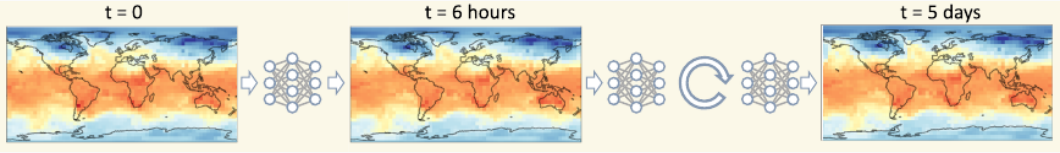
The mixture weight reflects the member's relative contributions to predictive skill over a training period.

For example, for temperature forecasting, the BMA mixture components can be chosen as Gaussian with a linearly bias-corrected mean, and so the BMA forecast density is

$$y \mid x_1, \ldots, x_M \sim \sum_{m=1}^{M} w_m \mathcal{N}(c_{0m} + c_{1m} x_m, \sigma^2),$$

with BMA weights, $w_1, \ldots, w_M$, that are nonnegative and sum to 1, bias parameters $c_{01}, \ldots, c_{0M}$ and $c_{11}, \ldots, c_{1M}$, and a common variance, σ^2.

AI-based data-driven weather prediction as a new paradigm



Rasp et al. (2024)

Potential benefits of AI weather models:

- better forecasts than physics-based weather models?
- potential of skillful forecasts at time scales or in locations where physical models have little or no skill
- substantially lower computational costs and faster generation of forecasts (once trained)

Do AI weather models produce better forecasts than physics-based models?

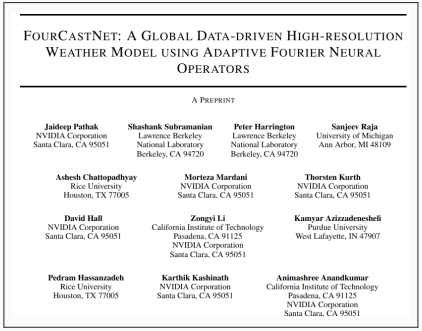
... probably (depending on the definition of *better*) ...

"History" of data-driven weather prediction

- early attempts (ca. 2018): proof of concept, but forecast skill is only comparable to simple toy models
- incremental progress, but forecast skill remains far behind that of operational physics-based models
- consensus (ca. 2021): challenging operational physics-based weather models seems highly unlikely for the foreseeable future

The AI revolution of weather prediction?

Unprecedented and surprising developments since February 2022:



FOURCASTNET: A GLOBAL DATA-DRIVEN HIGH-RESOLUTION WEATHER MODEL USING ADAPTIVE FOURIER NEURAL OPERATORS

A PREPRINT

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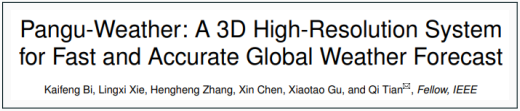
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Februar 2022: FourCastNet (Nvidia)



Pangu-Weather: A 3D High-Resolution System for Fast and Accurate Global Weather Forecast

Kaifeng Bi, Lingxi Xie, Hengheng Zhang, Xin Chen, Xiaotao Gu, and Qi Tian✉, *Fellow, IEEE*

November 2022: Pangu-Weather (Huawei)



GraphCast: Learning skillful medium-range global weather forecasting

Remi Lam[*,1], **Alvaro Sanchez-Gonzalez**[*,1], **Matthew Willson**[*,1], **Peter Wirnsberger**[*,1], **Meire Fortunato**[*,1], **Alexander Pritzel**[*,1], **Suman Ravuri**[1], **Timo Ewalds**[1], **Ferran Alet**[1], **Zach Eaton-Rosen**[1], **Weihua Hu**[1], **Alexander Merose**[2], **Stephan Hoyer**[2], **George Holland**[1], **Jacklynn Stott**[1], **Oriol Vinyals**[1], **Shakir Mohamed**[1] and **Peter Battaglia**[1]

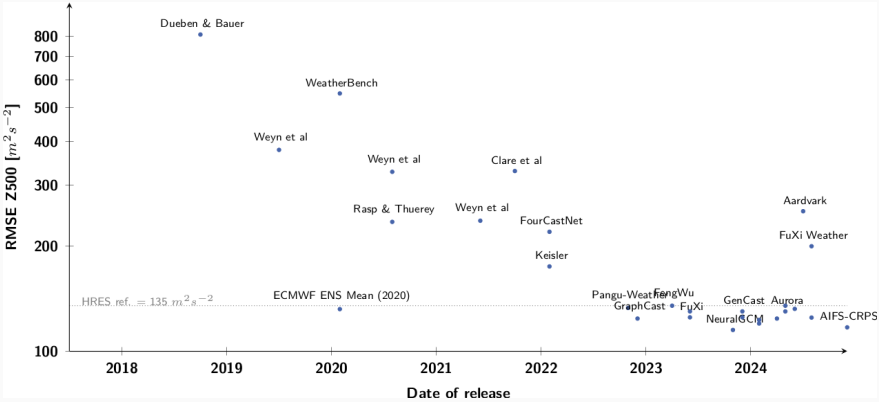
[*]equal contribution, [1]DeepMind, [2]Google

December 2022: GraphCast (Google/Deepmind)

... and more: ClimaX (Microsoft Research, January 2023), FengWu (Shanghai AI Laboratory, April 2023), FuXi (Fudan University, June 2023), ..., AIFS (ECMWF, June 2024), ...

Data-driven weather models: Successes

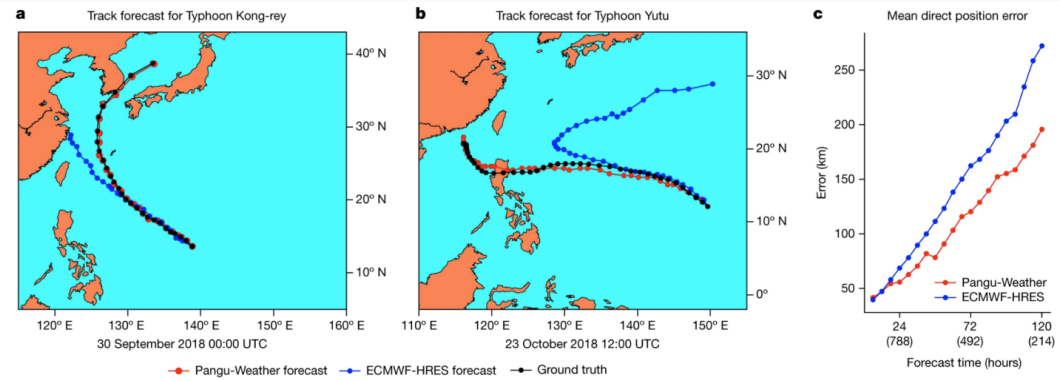
All these AIWP models reach or surpass the performance of state-of-the-art operational physical weather models ...



based on data collected by Stephan Rasp

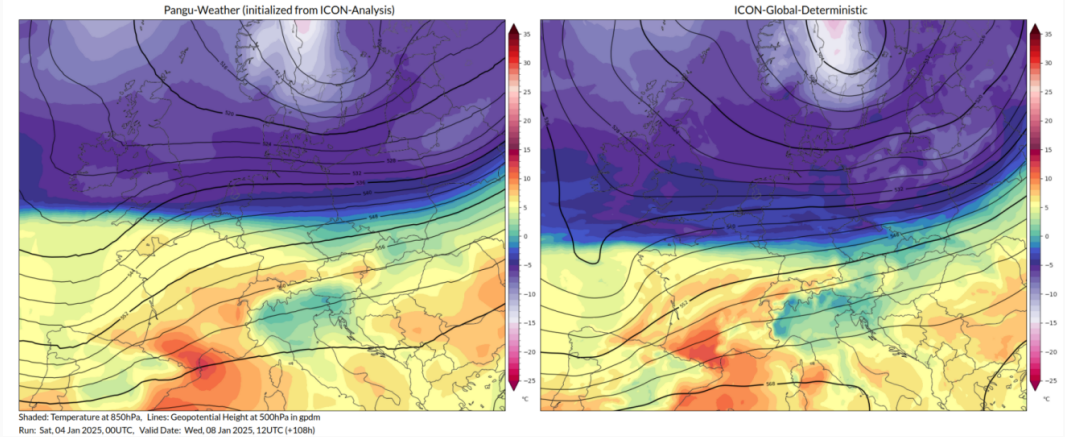
Data-driven weather models: Successes

... including predictions of complex phenomenons such as tropical cyclones



Bi et al. (2023), Nature

Exemplary Pangu-Weather forecasts



(Electricity) costs for one year of forecasts[1]

Physics-based NWP ensemble model



≈ 440 000 €

AI-based (Pangu-Weather) ensemble model



≈ 200 €

Training: ≈ 10 000 €

[1]Based on experiments on HoReKa for electricity costs of 50 Cent/kWh, 40 ensemble members, 2 models runs per day for 365 days. Images: generated by DALL-E 3 via Bing AI Copilot; lego.com

Climate forecasting

- Goal: Predict or project the future state and statistics of the Earth system (temperature, precipitation, extremes, etc.) over months to centuries:
  - Seasonal: months
  - Decadal: years–decades
  - Long-term: decades–centuries
- Approach:
  - Run a coupled Earth-system model (ESM) forward in time
  - Use ensembles to represent internal variability and model uncertainty
  - For long-term projections, specify future external forcings (e.g., greenhouse-gas and aerosol emissions)
- Outputs:
  - Probabilistic predictions of means, trends, and extremes
  - Climate indices (e.g., ENSO, NAO, PDO) and impact-relevant quantities

Weather vs. climate forecasting

- typical lead time: hours–weeks vs. months–centuries
- primary objective: predict the atmospheric state at a specific time vs. predict climate statistics and their evolution
- initialization: detailed atmospheric state vs. coupled Earth-system state; importance depends on lead time
- key physical processes: fast atmospheric dynamics vs. interactions between atmosphere, ocean, land, ice, and biosphere
- resolution: $\sim$1–10 km vs. $\sim$50–200 km
- typical users: aviation, emergency management, energy trading, ... vs. policy makers, impact modelers, long-term planners, ...

Challenges in climate forecasting

- Predictability: chaotic dynamics and intrinsic variability limit predictability, depending on the variable, spatial scale, and horizon
- Model uncertainty: unresolved physical processes (e.g., clouds, convection, ocean mixing) and imperfect parameterizations
- Initialization: incomplete knowledge of the current state of the coupled Earth system
- Forcing uncertainty: future greenhouse-gas emissions, aerosols, land use, and other external drivers are unknown
- Extremes: rare and potentially compound events are difficult to simulate and predict
- Evaluation: limited observations and few independent forecast cases, particularly for long lead times and extremes
- Uncertainty quantification: representing and communicating the different sources of uncertainty

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1. Introduction, Examples and Applications

1.1 Probabilistic and point forecasts

1.2 Examples and applications

1.2.1 Economic forecasting

1.2.2 Political and social science

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Examples of environmental sciences applications and prediction targets

- Hydrology and water resources: river discharge, water levels, soil moisture, ...
- Air quality and atmospheric composition: pollutants such as PM2.5, ozone etc.
- Wildfires: ignition probability, fire occurrence, burned area, fire spread, smoke exposure, ...
- Other environmental hazards: earth quakes, landslides, avalanches, ...
- Oceanography and marine ecosystems: ocean currents, sea level, waves, sea ice, harmful algal blooms, ...
- Agriculture: crop yield, soil moisture, plant diseases, pest outbreaks
- Biodiversity and ecological dynamics: species abundance, occupancy, migration, population dynamics, ...

Examples of environmental sciences applications and prediction targets

These fields strongly differ in terms of the quantities of interest, forecast horizons, and the degree of maturity.

Forecasting methods and approaches often include combinations of

- process-based models (e.g., physics-based hydrological models or biological models of plants)
- data-driven approaches relying on observation data (e.g., measurements, satellite imagery, weather forecasts, climate model projections, ...)
- statistical and machine learning methods (time series models, Bayesian statistics, post-processing, spatio-temporal statistics, ...)
- hybrid approaches combining process-based and data-driven modeling

Hydrological forecasting

Floods are among the most dangerous natural hazards in Europe.

For example, the flooding in the Ahr valley and at the Erft in 2021 caused at least 188 deaths and economic losses of more than €40 billion.



https://www.bundeswirtschaftsministerium.de/Redaktion/DE/Downloads/M-O/Merkblaetter/merkblatt-klimawandelfolgen-in-deutschland-05.pdf

Hydrological forecasting

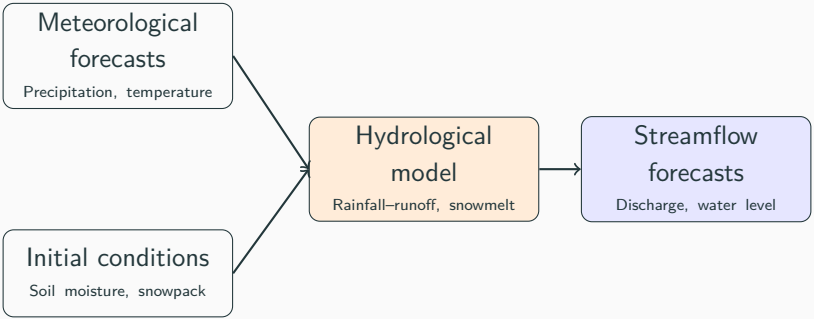
The absence of water can also cause severe economic problems.

In 2026, rivers in Germany experienced extremely low water levels, with economic damages estimated to be up to €2 billion, and decreasing GPD growth by up to 0.5 percentage points.



https://www.bafg.de/DE/5_Informiert/2_Publikationen/Niedrigwasserbericht/niedrigwasserbericht_node.html

Hydrological forecasting



Hydrological models

Conceptual water balance models (e.g., LARSIM) are often used by the flood forecasting centres. They divide the river basins into small computational units and simulate the main processes of the terrestrial water cycle, and the routing of water through the river network.

Observed meteorological data (station measurements and radar-based precipitation) drive the model up to the present. From there, numerical weather predictions, drive the model into the future and yield forecasts of discharge and water level.

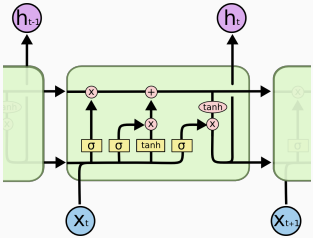
To account for uncertainties, these models are often run as ensembles, e.g., by utilizing ensemble weather forecasts as inputs.

Details and overviews: Cloke and Pappenberger (2009) and Troin et al. (2021).

Hydrological models

Over the past years, Long Short-Term Memory (LSTM) networks turned out to be exceptionally well suited for rainfall-runoff modeling.

Trained jointly on hundreds of catchments, LSTMs learn hydrological behavior that generalizes across basins and outperform established conceptual and process-based models.

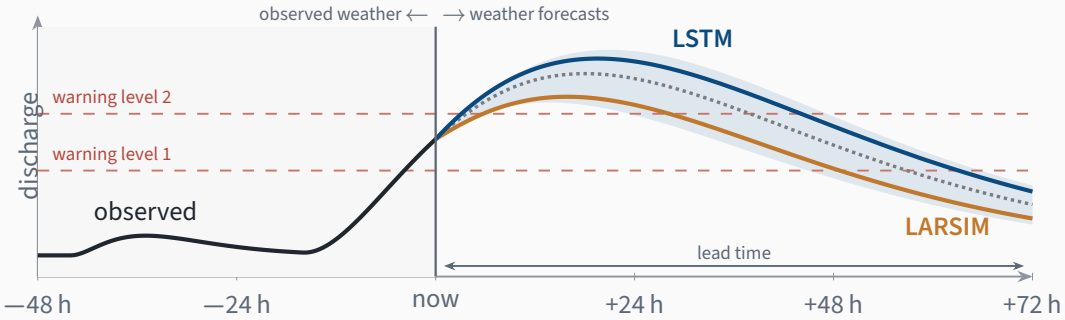


https://colah.github.io/posts/2015-08-Understanding-LSTMs

Google's Flood Hub issues operational LSTM-based flood forecasts in more than 80 countries.

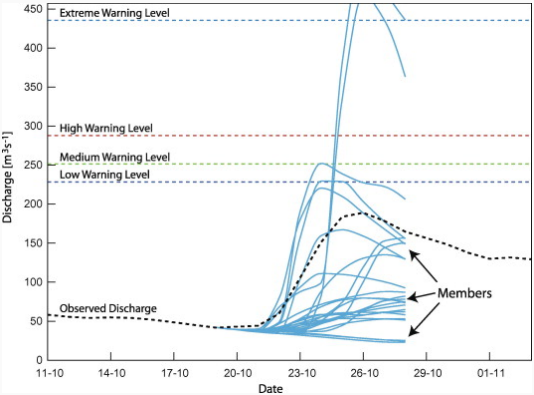
Details: Kratzert et al. (2018) and Nearing et al. (2024).

Hydrological forecasting



Hydrological ensemble models

Ensemble forecasts can be generated by using different hydrological models or ensemble weather forecasts



Cloke and Pappenberger (2009)

Post-processing can be applied to the ensemble weather forecasts used as inputs, but also to the hydrological ensemble forecasts obtained as output of the hydrological model.

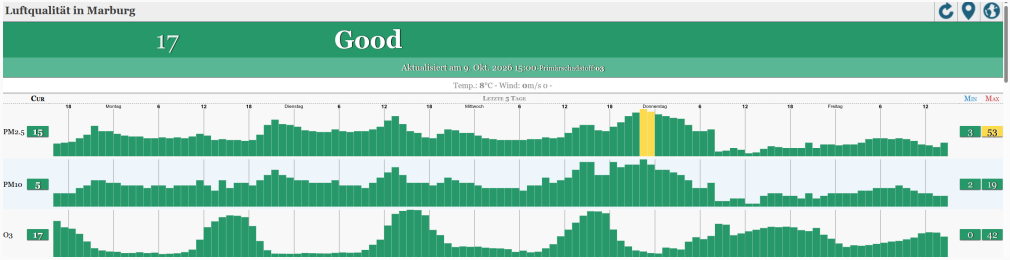
Advertisement

Advanced software lab / project work in the current winter semester: Development of an LSTM model for flood prediction in Hesse, together with HLNUG.

Application is possible until October 25, 2026.



Air quality forecasting



https://aqicn.org/city/germany/hesse/marburg/de/m/

Central question: How likely is the concentration of a pollutant to exceed a health-relevant threshold, at a particular location and time? Target variables:

- particulate matter (PM10, PM2.5) from traffic, industry, domestic heating, wildfires
- gaseous pollutants (O_3, NO_2, SO_2, CO) from atmospheric chemical reactions (ozone), traffic, combustion, and industry (others)
- smoke and dust from wildfires or Sahara dust episodes

Air quality forecasting

Emissions and observations

Traffic, industry, fires; monitoring stations and satellites

↓

Meteorological forecasts

Wind, temperature, boundary-layer mixing, precipitation

↓

Chemical transport model

Transport, dispersion, chemical reactions, deposition

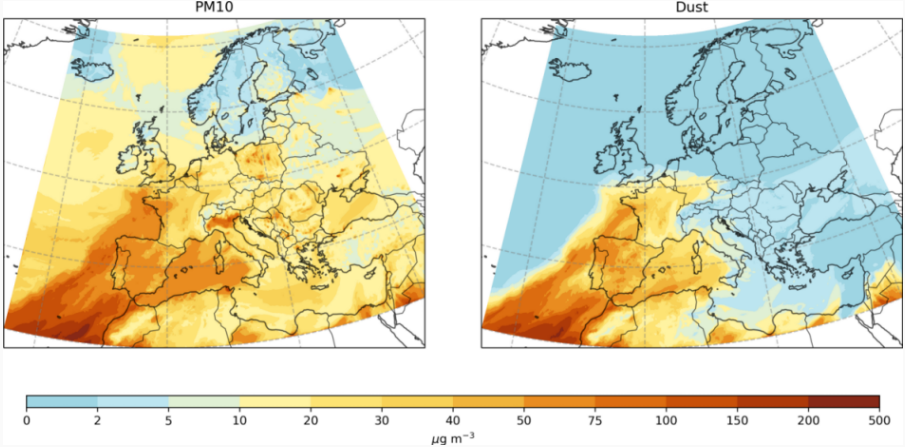
↓

Forecast and uncertainty estimation

Multi-model ensemble, statistical post-processing, predictive distributions

Air quality forecasting

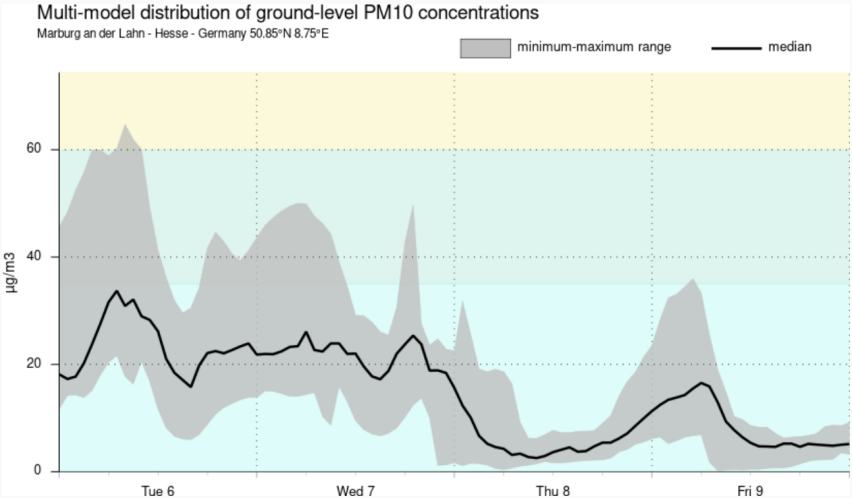
The Copernicus Atmosphere Monitoring Service (CAMS) provides regional air-quality ensemble forecasts from 11 chemical transport models



https://atmosphere.copernicus.eu/cams-air-boreal-winter-2324

Air quality forecasting

Real-time forecasts: https://charts.ecmwf.int/catalogue/packages/cams_air_quality_eea/



Overview

1. Introduction, Examples and Applications

1.1 Probabilistic and point forecasts

1.2 Examples and applications

1.3 Forecasting competitions and prediction markets

Forecasting competitions

Structured, open experiments in which participants submit predictions for a common set of future outcomes, and performance is evaluated using a predefined metric.

Different types of competitions:

- domain-specific, statistical and machine-learning forecasting competitions
- general purpose predictive modelling competitions (e.g., on Kaggle), often focusing on classification or regression rather than forecasting future events
- human geopolitical and current-events forecasting tournaments
  - examples: Metaculus, Good Judgement Project[2]

[2] see also Tetlock, P.E. (2005) *Expert Political Judgment: How Good Is It? How Can We Know?*

Forecasting competitions: Examples

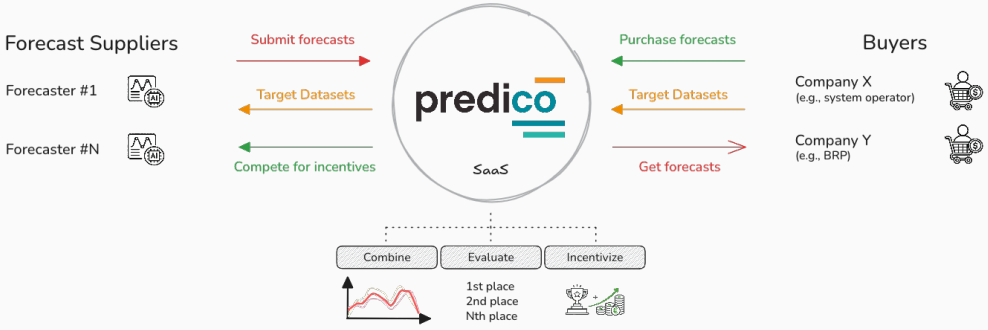
- Global Energy Forecasting Competition (GEFCOM) 2012, 2014, 2017 with a focus on load, electricity price, wind power and solar power forecasting
- Makridakis (or M-)Competitions since 1982 (latest edition M6 Competition 2022) for time series forecasting
- WMO Sub-seasonal to Seasonal (S2S) AI Challenge on sub-seasonal weather prediction

Real-time forecasting competitions:

- Survey of Professional Forecasters, macroeconomic forecasting
- COVID-19 Forecasting Hub, epidemiology, `https://covid19forecasthub.org/`
- ECMWF AI Weather Quest, subseasonal weather prediction, `https://aiweatherquest.ecmwf.int/`

Collaborative forecasting platforms with market structures

Platforms such as Predico[3] are multilateral, collaborative markets for forecasts, connecting entities with common interests to procure and sell forecasts.

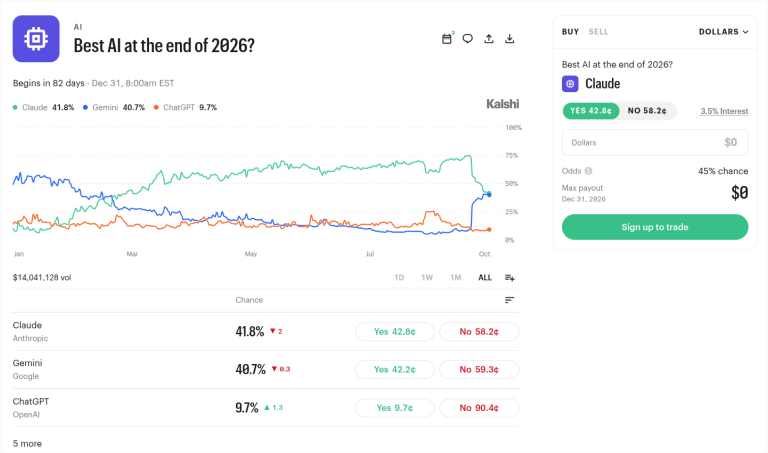


https://predico.inesctec.pt/docs/

[3] https://innovation.eliagroup.eu/en/projects/predico-collaborative-forecasting-platform

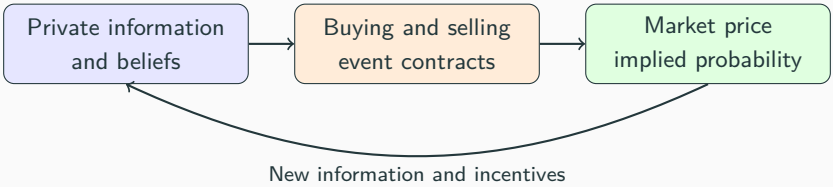
Prediction markets

Prediction markets allow participants to buy and sell contracts whose payoffs depend on the outcome of a future event. The market price aggregates information through participants' willingness to trade.



https://kalshi.com/markets/kxllm1/yearend-top-llm/kxllm1-26dec31

Prediction markets



- A binary contract pays 1 if an event occurs and 0 otherwise.
- Its price can be interpreted as an implied probability.
- Trading aggregates information, but prices need not be calibrated.
- Examples: Polymarket, Kalshi, Manifold Markets.

Prediction markets

While proponents argue that prediction markets aggregate information efficiently to generate accurate real-time forecasts, these markets raise several major ethical concerns:

- insider trading and manipulation without a regulatory framework
- incentives to cause predicted events
- gambling under the guise of finance
- gamification and aggressive social media marketing of prediction platforms disproportionately targets younger traders
- profiting from misfortune and human suffering by betting on negative events
- predictions might change outcomes in reality, e.g., panic selling in stock markets or alterations in political strategy

Further reading: `https://cer.econ.columbia.edu/news/use-caution-hidden-risks-prediction-markets`

Insider trading on prediction markets

- Between 30 December 2025 and 2 January 2026, a US special forces soldier places a total of $32,500 on bets on Venezuela's President Nicolás Maduro being out of office by the end of January 2026.

  When Maduro is seized by US special forces on 3 January 2026, the soldier wins $436,000.

- Surrounding the US strikes on Iran in February 2026, six accounts created on Polymarket in February win $1.2 million in wagers on a US strike on Iran happening by 28 February.

  One of those accounts later made $163,000 by correctly betting on a US-Iran ceasefire by 7 April.



https://www.bbc.com/news/articles/cge0grppe3po

2. Prediction Spaces, Calibration and Sharpness

Disclaimer

We will now turn to a formal, measure-theoretic introduction of prediction spaces, which are probability spaces tailored to the study of forecasting.

We will then discuss the critical notions of calibration, dispersion, and sharpness, and argue that the goal in probabilistic forecasting is to maximize the sharpness of the predictive distributions, subject to calibration.

The exposition largely follows Gneiting and Ranjan (2013).

In applications, the measure-theoretic construction will remain hypothetical.

Overview

2. Prediction Spaces, Calibration and Sharpness

2.1 Prediction spaces

2.2 Calibration

2.3 Sharpness

Prediction spaces

Consider the joint distribution of multiple probabilistic forecasts and the observation on a probability space

$$(\Omega, \mathcal{A}, \mathbb{Q}).$$

We assume that the elements of the sample space Ω can be identified with tuples of the form

$$(P_1, \ldots, P_k, Y),$$

where each of $P_1, \ldots, P_k$ is a probability measure on the outcome space for the observation, Y,

$$(\Omega_Y, \mathcal{A}_Y).$$

The probability measure $\mathbb{Q}$ on $(\Omega, \mathcal{A})$ specifies the joint distribution of the probabilistic forecasts and the observation.

Information sets

For $i = 1, \ldots, k$, we require the random probability measure P_i to be measurable with respect to the sub-σ-algebra $\mathcal{A}_i \subseteq \mathcal{A}$ that encodes the forecast's information set:

- consists of data, expertise, theories and assumptions at hand

The probabilistic forecasts $P_1, \ldots, P_k$ might stem from distinct experts, organizations or statistical models, as commonly encountered in the practice of forecasting.

Notation

In what follows in the present section,

- we restrict the discussion to the case of a real-valued observation, so that $(\Omega_Y, \mathcal{A}_Y) = (\mathbb{R}, \mathcal{B})$
- a probabilistic forecast then corresponds to (a Lebesgue-Stieltjes probability measure on the real line, which we identify with the associated) right-continuous CDF
- We write $\sigma(\mathcal{A}_1, \ldots, \mathcal{A}_m)$ and $\sigma(X_1, \ldots, X_n)$ to denote the σ-algebra generated by the families $\mathcal{A}_1, \ldots, \mathcal{A}_m$ of subsets of Ω, or by the random variables $X_1, \ldots, X_n$, respectively.
- We use the symbol $\mathcal{L}$ to denote an unconditional or conditional distribution.

Prediction space

Definition 2.1: Prediction space

Let $k \geq 1$ be an integer. A **prediction space** is a probability space $(\Omega, \mathcal{A}, \mathbb{Q})$ together with sub-σ-algebras $\mathcal{A}_1, \ldots, \mathcal{A}_k \subseteq \mathcal{A}$, where the elements of the sample space Ω can be identified with tuples

$$(F_1, \ldots, F_k, Y, V)$$

such that

(P1) for $i = 1, \ldots, k$, F_i is a CDF-valued random quantity that is measurable with respect to the sub-σ-algebra $\mathcal{A}_i$,

(P2) Y is a real-valued random variable,

(P3) V is a random variable that is uniformly distributed on the unit interval and independent of $\mathcal{A}_1, \ldots, \mathcal{A}_k$ and Y.

Ideal probabilistic forecasts

Subsequent definitions and phrases such as *almost surely* or *with positive probability* refer to the probability measure $\mathbb{Q}$ on $(\Omega, \mathcal{A})$ that determines the joint distribution of the probabilistic forecasts and the observations.

The sub-σ-algebra $\mathcal{A}_i$ encodes the information set for the CDF-valued random quantity F_i which may, but need not, be ideal in the following sense.

Definition 2.2

The CDF-valued random quantity F_i is **ideal** relative to the sub-σ-algebra $\mathcal{A}_i$ if

$$F_i = \mathcal{L}(Y \mid \mathcal{A}_i)$$

almost surely.

Example: Binary events

We consider probability forecasts of a dichotomous outcome, such as rain *vs.* no rain, or recession *vs.* no recession.

We identify a success or occurrence with the outcome $Y = 1$, and a non-success with $Y = 0$.

The CDF-valued random quantity F_i, where $i = 1, \ldots, k$, thus is of the form

$$F_i(y) = (1 - p_i)\mathbf{1}\{y \geq 0\} + p_i\mathbf{1}\{y \geq 1\}$$

and can be identified with the probability forecast p_i for a success, where

$$p_i \in [0, 1]$$

is a random variable.

Example: Density forecasts

Typically, it suffices to consider the joint distribution of the tuple $(F_1, \ldots, F_k, Y)$, without any need to explicitly specify other facets of the prediction space.

To define a prediction space, let

$$Y \mid \mu \sim \mathcal{N}(\mu, 1) \qquad \text{where} \qquad \mu \sim \mathcal{N}(0, 1),$$

and let τ attain the values 1 and -1 with equal probability, independently of μ and Y, i.e.,

$$\mathbb{P}(\tau = 1) = \mathbb{P}(\tau = -1) = \frac{1}{2}.$$

Quiz

In this example, a forecaster has access to the value of μ. Which predictive distribution is ideal relative to the σ-algebra $\sigma(\mu)$ generated by μ?

A: $\mathcal{N}(\mu, 1)$

B: $\mathcal{N}(0, 2)$

C: $\mathcal{N}(\mu, 2)$

D: $\mathcal{N}(0, 1)$

Example: Density forecasts

Consider the following four density forecasts:

Forecast	Predictive Distribution
Perfect	$F_1 = \mathcal{N}(\mu, 1)$
Climatological	$F_2 = \mathcal{N}(0, 2)$
Unfocused	$F_3 = \frac{1}{2}\left(\mathcal{N}(\mu, 1) + \mathcal{N}(\mu + \tau, 1)\right)$
Sign-reversed	$F_4 = \mathcal{N}(-\mu, 1)$

- The perfect forecast is ideal relative to the sub-σ-algebra generated by μ.
- The climatological forecast is ideal relative to the trivial sub-σ-algebra.
- The unfocused and sign-reversed forecasts are not ideal.

Details and further properties: Exercises

Overview

2. Prediction Spaces, Calibration and Sharpness

2.1 Prediction spaces

2.2 Calibration

2.3 Sharpness

Calibration: Motivation

Intuitively, a probabilistic forecast is calibrated if it is statistically consistent with the observations:

The observation should look like a random draw from the predictive distribution.

Example: If a weather service issues a 30% chance of rain on 100 days, we expect rain on about 30 of these days.

Calibration is a property of the joint distribution of forecasts and observations, and can only be assessed from a sample of forecast cases.

In the following, we introduce several notions of calibration and dispersion, and diagnostic tools for checking them (Gneiting et al., 2007; Gneiting and Ranjan, 2013).

Overview

2. Prediction Spaces, Calibration and Sharpness

2.1 Prediction spaces

2.2 Calibration

2.2.1 Probability integral transform

2.2.2 Notions of calibration and dispersion

2.2.3 Tools for calibration checks

2.2.4 Probability forecasts of a binary event

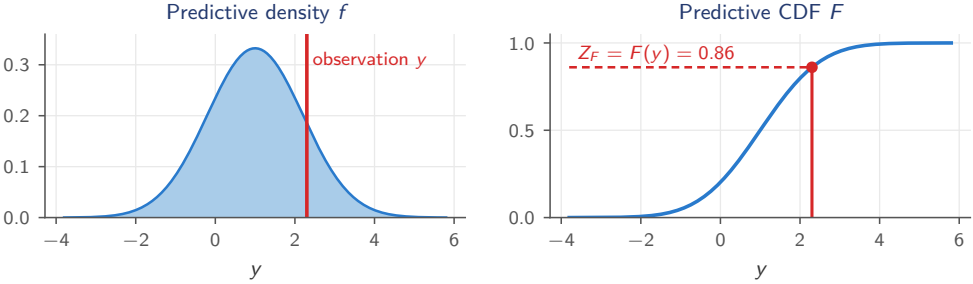
2.3 Sharpness

Probability integral transform

If F denotes a fixed, non-random predictive CDF for an observation Y, the probability integral transform (PIT) is the random variable

$$Z_F = F(Y),$$

i.e., the value that the predictive CDF attains at the observation.



Probability integral transform

It is well known that if F is continuous and $Y \sim F$, then Z_F is standard uniform.

Proof (for strictly increasing F): For $z \in (0, 1)$,

$$\mathbb{Q}(Z_F \leq z) = \mathbb{Q}(F(Y) \leq z) = \mathbb{Q}(Y \leq F^{-1}(z)) = F(F^{-1}(z)) = z.$$

However: If F has jumps, then $F(Y)$ is not uniform, even if $Y \sim F$.

Example: If $Y \sim \text{Ber}(p)$, then $F(Y)$ only attains the values $1 - p$ and 1.

Predictive distributions with jumps are common in practice, e.g., for

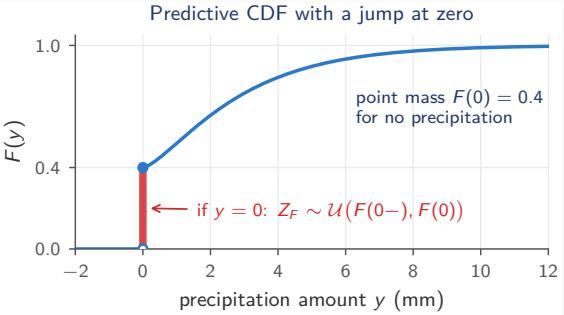
- count data (e.g., the number of goals in a soccer match),
- precipitation amounts (point mass at zero),
- ensemble forecasts, which are identified with their empirical CDF.

Randomized probability integral transform

Let V be standard uniform and independent of Y, and write $F(y-) = \lim_{x \uparrow y} F(x)$ for the left-hand limit of F at $y \in \mathbb{R}$. The randomized PIT

$$Z_F = F(Y-) + V\,(F(Y) - F(Y-))$$

draws a uniform value from the jump of F at the observation (if there is one).



If $Y \sim F$, then the randomized PIT is standard uniform for any fixed CDF F (Rüschendorf, 2009).

Proof: Exercise

PIT in the prediction space setting

In the prediction space setting, F is a CDF-valued random quantity, and we use the auxiliary variable V from assumption (P3) in Definition 2.1.

Definition 2.3: Probability integral transform

In the prediction space setting, the random variable

$$Z_F = F(Y-) + V\,(F(Y) - F(Y-))$$

is the **probability integral transform** (PIT) of the CDF-valued random quantity F.

In a nutshell, the PIT is the value that the predictive CDF attains at the observation, with suitable adaptions at any points of discontinuity.

The PIT takes values in the unit interval, so that $\mathrm{Var}(Z_F) \in [0, \frac{1}{4}]$.

Quiz

The predictive distribution is $F = \mathcal{N}(0, 1)$, and we observe $y = 1.645$. What is the value of the PIT?

A: 0.05

B: 0.5

C: 0.95

D: 1.645

Overview

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Notions of calibration

In what follows, we use the terms CDF-valued random quantity and forecast interchangeably.

Definition 2.4: Calibration

In the prediction space setting, let F be a forecast with PIT Z_F.

(a) F is **marginally calibrated** if $\mathbb{E}_{\mathbb{Q}}[F(y)] = \mathbb{Q}(Y \leq y)$ for all $y \in \mathbb{R}$.

(b) F is **probabilistically calibrated** if its PIT Z_F is uniformly distributed on the unit interval.

- In the definition of marginal calibration, the left-hand side depends on the law of the forecast, whereas the right-hand side depends on the law of the observation ("forecast climate = actual climate").
- Probabilistic calibration involves the joint law of the forecast and the observation.

Definition 2.5: Dispersion

In the prediction space setting, let F and G be forecasts with PITs Z_F and Z_G.

(a) F is **overdispersed** if $\mathrm{Var}(Z_F) < \frac{1}{12}$, **neutrally dispersed** if $\mathrm{Var}(Z_F) = \frac{1}{12}$, and **underdispersed** if $\mathrm{Var}(Z_F) > \frac{1}{12}$.

(b) F is **at least as dispersed** as G if $\mathrm{Var}(Z_F) \leq \mathrm{Var}(Z_G)$, and **more dispersed** than G if $\mathrm{Var}(Z_F) < \mathrm{Var}(Z_G)$.

(c) F is **regular** if the support of the distribution of Z_F is the unit interval.

Intuition: Recall that $\frac{1}{12}$ is the variance of the uniform distribution on $[0, 1]$. If F is too narrow (underdispersed), the observation often falls into its tails, so that PIT values close to 0 or 1 are frequent and $\mathrm{Var}(Z_F)$ is large.

Calibration and dispersion

Dispersion is defined via the PIT and thus involves the joint law of the forecast and the observation. In contrast, the spread of the predictive distribution itself is a measure of sharpness that does not consider the observation (Section 2.3).

Lemma 2.6

A probabilistically calibrated forecast is neutrally dispersed and regular.

The converse is not necessarily true, in that a forecast which is neutrally dispersed need not be calibrated nor regular.

However, an ideal forecast is always calibrated.

Quiz

For a large sample of forecast cases, the PIT values of a forecasting method have an empirical variance of 0.12. The forecasts are

A: overdispersed

B: neutrally dispersed

C: underdispersed

D: probabilistically calibrated

Ideal forecasts are calibrated

Theorem 2.7

A forecast that is ideal relative to a σ-algebra is both marginally calibrated and probabilistically calibrated.

Proof: Let $F = \mathcal{L}(Y \mid \mathcal{A}_0)$ be ideal relative to $\mathcal{A}_0$, i.e., $F(y) = \mathbb{Q}(Y \leq y \mid \mathcal{A}_0)$ almost surely for all $y \in \mathbb{R}$.

Marginal calibration: By the tower property of conditional expectations,

$$\mathbb{E}_{\mathbb{Q}}[F(y)] = \mathbb{E}_{\mathbb{Q}}\left[\mathbb{Q}(Y \leq y \mid \mathcal{A}_0)\right] = \mathbb{E}_{\mathbb{Q}}\left[\mathbb{E}_{\mathbb{Q}}\left[\mathbf{1}\{Y \leq y\} \mid \mathcal{A}_0\right]\right] = \mathbb{Q}(Y \leq y).$$

Probabilistic calibration: Given $\mathcal{A}_0$, the forecast F is fixed and $Y \sim F$. By the uniformity of the randomized PIT for fixed CDFs, $\mathbb{Q}(Z_F \leq z \mid \mathcal{A}_0) = z$ for $z \in (0, 1)$, and so

$$\mathbb{Q}(Z_F \leq z) = \mathbb{E}_{\mathbb{Q}}\left[\mathbb{Q}(Z_F \leq z \mid \mathcal{A}_0)\right] = z.$$

■

Quiz

Recall the four density forecasts (perfect, climatological, unfocused, sign-reversed). We simulate 10 000 forecast cases and plot a histogram of the PIT values for each forecast. For which forecasts do we expect a uniform PIT histogram?

A: only the perfect forecast

B: the perfect and the climatological forecasts

C: the perfect, the climatological and the unfocused forecasts

D: all four forecasts

Example: Density forecasts (continued)

Recall that $Y \mid \mu \sim \mathcal{N}(\mu, 1)$ with $\mu \sim \mathcal{N}(0, 1)$, and $\mathbb{Q}(\tau = 1) = \mathbb{Q}(\tau = -1) = \frac{1}{2}$:

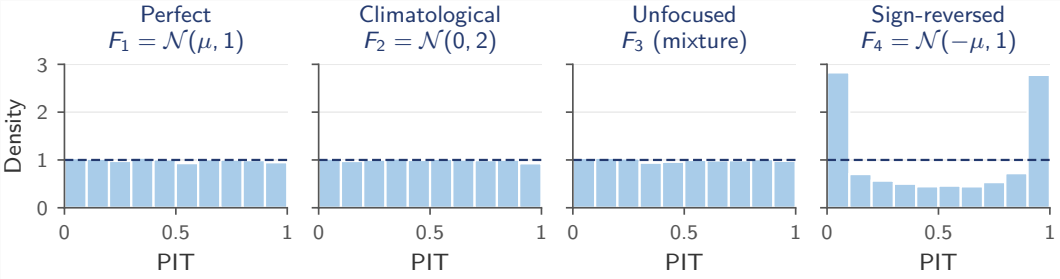
Forecast	Predictive distribution	Marginally calibrated	Probabilistically calibrated	Ideal
Perfect	$F_1 = \mathcal{N}(\mu, 1)$	Yes	Yes	Yes
Climatological	$F_2 = \mathcal{N}(0, 2)$	Yes	Yes	Yes
Unfocused	$F_3 = \frac{1}{2}\left(\mathcal{N}(\mu, 1) + \mathcal{N}(\mu + \tau, 1)\right)$	No	Yes	No
Sign-reversed	$F_4 = \mathcal{N}(-\mu, 1)$	Yes	No	No

The perfect and the climatological forecast are ideal, and thus calibrated by Theorem 2.7.

The unfocused forecast is probabilistically but not marginally calibrated, and the sign-reversed forecast is marginally but not probabilistically calibrated. Hence, there is no information set relative to which they are ideal (proofs: Exercises).

Example: Density forecasts (continued)

PIT histograms for a simulated sample of size 10 000:



The PIT histogram cannot distinguish between the perfect, the climatological and the unfocused forecast.

→ We need additional criteria: sharpness (Section 2.3) and proper scoring rules (Section 3).

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PIT histograms

Probabilistic calibration is considered the most critical requirement for probabilistic forecasts that take the form of predictive CDFs.

Checks for the uniformity of the PIT have formed a cornerstone of density forecast evaluation (Dawid, 1984; Diebold et al., 1998).

In practice, one observes a sample

$$\{(F_{1j}, \ldots, F_{kj}, y_j) : j = 1, \ldots, J\}$$

from the joint distribution of the probabilistic forecasts and the observation, and the uniformity of the PIT is assessed empirically.

The prevalent way of doing this is by plotting histograms of the PIT values for the various forecasting methods.

Interpreting PIT histograms

For a probabilistically calibrated forecast, the PIT histogram is uniform. Typical deviations from uniformity are:

- U-shaped: underdispersed predictive distributions, with prediction intervals that are too narrow on average
- hump-shaped (inverse U-shaped): overdispersed predictive distributions
- skewed: predictive distributions that are biased in their location

Example: Let $Y = X + \epsilon$, where X and ϵ are independent standard normal random variables, and consider the Gaussian forecast

$$F_\sigma = \mathcal{N}(X, \sigma^2).$$

Then F_σ is underdispersed if $\sigma < 1$, neutrally dispersed if $\sigma = 1$, and overdispersed if $\sigma > 1$. If $\sigma = 1$, then F_σ is ideal and thus calibrated.

Quiz

The PIT histogram of a forecasting method is decreasing from left to right, i.e., small PIT values are much more frequent than large ones. What does this indicate?

A: The predictive distributions are too narrow.

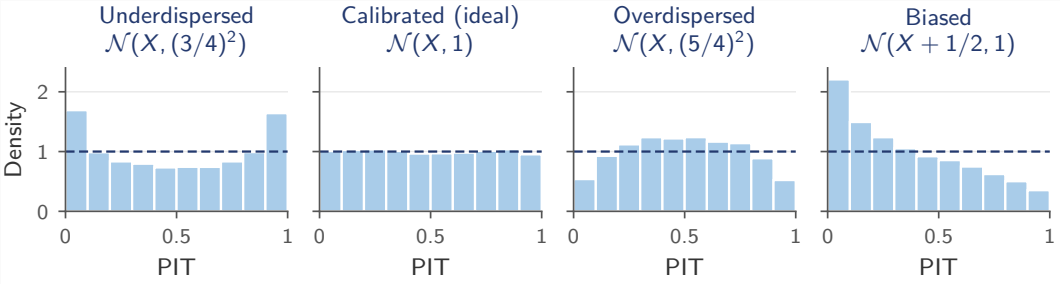
B: The predictive distributions are too wide.

C: The forecasts are systematically too high.

D: The forecasts are systematically too low.

Example: PIT histograms

PIT histograms for a simulated sample of size 10 000 from the joint distribution of Y and F_σ, and for a biased forecast $\mathcal{N}(X + \frac{1}{2}, 1)$:



Rank histograms for ensemble forecasts

Frequently, probabilistic forecasts take the form of a discrete sample $x_1, \ldots, x_m$, where each sample value is considered an equally likely realization of Y.

A prominent example are ensemble weather forecasts (Section 1.2.6). The sample can be identified with its empirical CDF,

$$F(y) = \frac{1}{m} \sum_{i=1}^{m} \mathbf{1}\{x_i \leq y\}.$$

In principle, the (randomized) PIT can be computed as usual.

However, if the sample values and the observation come from continuous distributions, there is another, preferred way of checking calibration based on ranks.

Rank histograms for ensemble forecasts

Let $x_{(1)} < \cdots < x_{(m)}$ denote the order statistics of $x_1, \ldots, x_m$, which divide the real line into $m+1$ bins. The rank of the observation y is

$$r = \#\{i \in \{1, \ldots, m\} : x_i \leq y\} + 1 \in \{1, \ldots, m+1\}.$$



If the sample values and the observation are exchangeable, the observation falls into any of the $m+1$ bins with equal probability $\frac{1}{m+1}$.

Ranks are composited over forecast cases and their empirical frequencies are plotted in a rank histogram (or Talagrand diagram), which is interpreted in the same way as a PIT histogram.

Quiz

An ensemble forecast has $m = 9$ members, and the ensemble members and the observation are exchangeable. With which probability does the observation fall outside the ensemble range $[x_{(1)}, x_{(m)}]$?

A: $\frac{1}{10}$

B: $\frac{1}{9}$

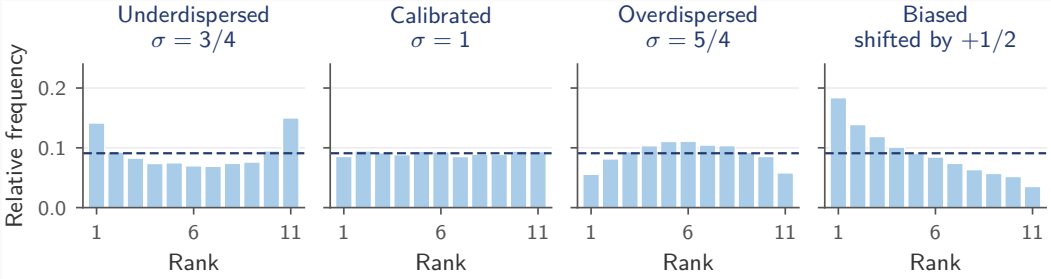
C: $\frac{2}{10}$

D: 0

Example: Rank histograms

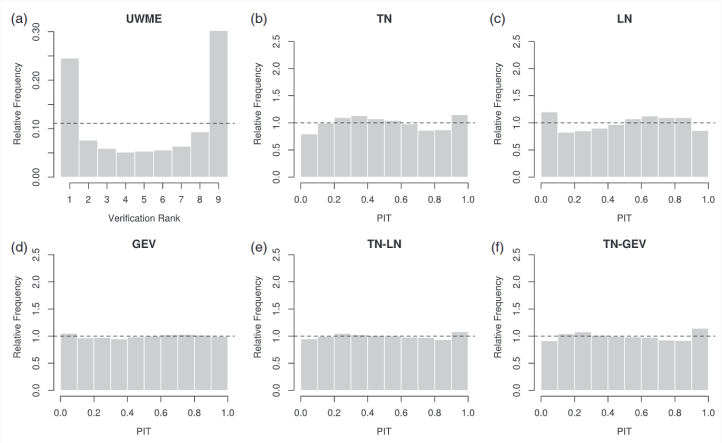
In particular, the ensemble range $[x_{(1)}, x_{(m)}]$ provides a central prediction interval with nominal coverage $\frac{m-1}{m+1}$.

Rank histograms for $m = 10$ members from $\mathcal{N}(X, \sigma^2)$ or $\mathcal{N}(X + \frac{1}{2}, 1)$, where $Y = X + \epsilon$ as before:



Example: Rank histograms

Raw ensemble weather forecasts are often underdispersed and biased. Interpretation and pitfalls: Hamill (2001). This can be corrected by post-processing:



Baran and Lerch (2016)

Calibration checks for time series forecasts

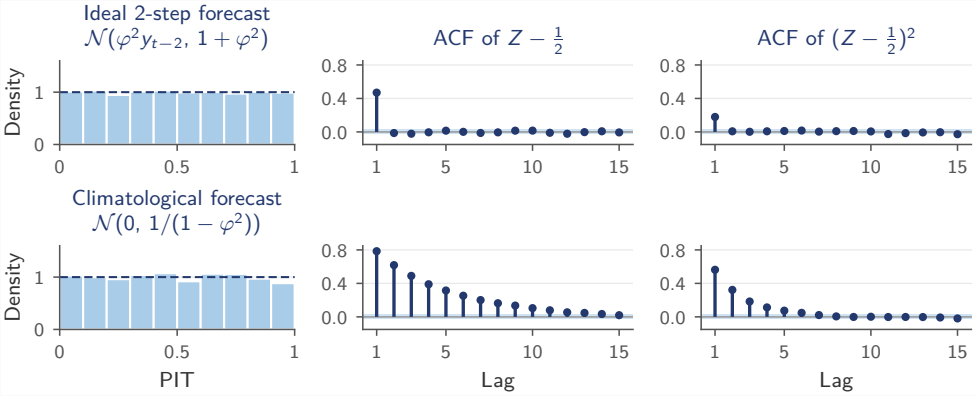
In time series settings, the observations are sequential, and the predictive distributions correspond to sequential k-step ahead forecasts.

The PIT values of ideal k-step ahead forecasts are independent at lags $\geq k$ (Diebold et al., 1998). Checks for uniformity can thus be supplemented by checks for independence, e.g., using sample autocorrelation functions (ACFs) of $Z - \frac{1}{2}$ and its powers.

Example: AR(1) process $Y_t = \varphi Y_{t-1} + \epsilon_t$ with $\varphi = 0.8$ and iid $\epsilon_t \sim \mathcal{N}(0, 1)$, and forecasts at lead time $k = 2$:

- ideal forecast given $y_{t-2}, y_{t-3}, \ldots$: $\mathcal{N}(\varphi^2 y_{t-2},\, 1 + \varphi^2)$
- climatological forecast: $\mathcal{N}\left(0,\, 1/(1 - \varphi^2)\right)$, ideal relative to the trivial σ-algebra

Example: 2-step ahead forecasts of an AR(1) process

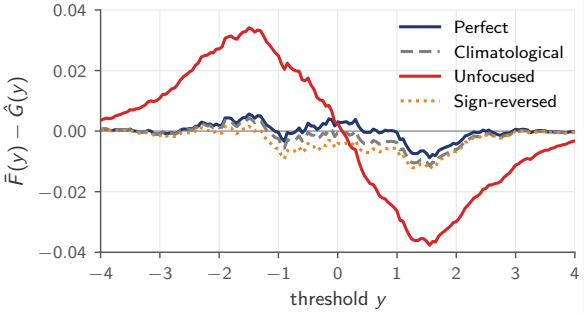


Both forecasts are probabilistically calibrated. The PIT values of the ideal forecast are correlated at lag 1 only, while those of the climatological forecast are strongly autocorrelated at lags ≥ 2: it ignores available information.

Checking marginal calibration

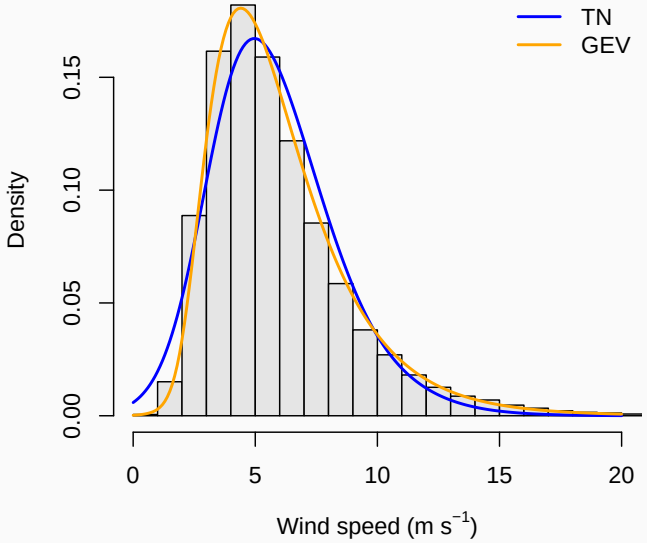
Marginal calibration concerns the equality of the forecast climate and the actual climate. To assess it, we can compare the average predictive CDF to the empirical CDF of the observations,

$$\bar{F}_J(y) = \frac{1}{J}\sum_{j=1}^{J} F_j(y) \qquad \text{vs.} \qquad \hat{G}_J(y) = \frac{1}{J}\sum_{j=1}^{J} \mathbf{1}\{y_j \le y\}.$$



Density forecasts example ($J = 10\,000$): only the unfocused forecast shows systematic deviations (cf. Gneiting et al., 2007).

Example: Choice of parametric forecast distribution in post-processing



based on data and models from Lerch and Thorarinsdottir (2013)

Overview

2. Prediction Spaces, Calibration and Sharpness

2.1 Prediction spaces

2.2 Calibration

2.2.1 Probability integral transform

2.2.2 Notions of calibration and dispersion

2.2.3 Tools for calibration checks

2.2.4 Probability forecasts of a binary event

2.3 Sharpness

Probability forecasts of a binary event

Recall that for a binary outcome $Y \in \{0, 1\}$, we identify the CDF-valued random quantity

$$F(y) = (1 - p)\mathbf{1}\{y \geq 0\} + p\mathbf{1}\{y \geq 1\}$$

with the probability forecast p for a success, i.e., $Y = 1$.

Definition 2.8: Conditional calibration

The probability forecast p is **conditionally calibrated** if

$$\mathbb{Q}(Y = 1 \mid p) = p \quad \text{almost surely.}$$

For example, among all cases in which a 30% chance of rain is forecast, it rains in 30% of the cases.

Calibration for binary events

If the outcome is binary, the notions of probabilistic calibration and conditional calibration are equivalent.

Theorem 2.9

Consider a prediction space with a binary outcome $Y \in \{0, 1\}$, and a forecast F that is identified with the probability forecast p. Then the following statements are equivalent:

(i) F is probabilistically calibrated, i.e., Z_F is uniformly distributed on the unit interval.

(ii) p is conditionally calibrated, i.e., $\mathbb{Q}(Y = 1 \mid p) = p$ almost surely.

(iii) F is ideal relative to the σ-algebra $\sigma(p)$ generated by p.

Proof idea

(ii) $\Leftrightarrow$ (iii) holds by definition, and (iii) $\Rightarrow$ (i) follows from Theorem 2.7. It remains to show that (i) $\Rightarrow$ (ii).

Let $q(x) = \mathbb{Q}(Y = 1 \mid p = x)$. Given $p = x$, the randomized PIT is

$$Z_F = \begin{cases} V(1-x) \sim \mathcal{U}(0, 1-x) & \text{if } Y = 0 \quad (\text{probability } 1 - q(x)), \\ 1 - x + Vx \sim \mathcal{U}(1-x, 1) & \text{if } Y = 1 \quad (\text{probability } q(x)). \end{cases}$$

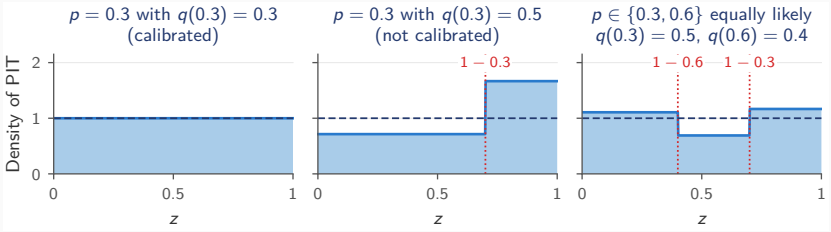
Hence, the conditional density of Z_F given $p = x$ is a step function with a jump at $1 - x$,

$$u(z \mid x) = \frac{1 - q(x)}{1 - x} \quad \text{for } z < 1 - x, \qquad u(z \mid x) = \frac{q(x)}{x} \quad \text{for } z > 1 - x,$$

and it is flat (i.e., uniform) if and only if $q(x) = x$.

Proof idea

Key idea: Suppose p takes finitely many values $x_1, \ldots, x_n \in (0,1)$ with probabilities $h_1, \ldots, h_n$. Then the density $u(z) = \sum_j h_j\, u(z \mid x_j)$ of Z_F is piecewise constant, with a jump of size $h_j\,(q(x_j) - x_j)/(x_j(1-x_j))$ at $z = 1 - x_j$.



If Z_F is uniform, there are no jumps, so $q(x_j) = x_j$ for all j, which is (ii). ■

General case: the same argument applies to arbitrary intervals of forecast values. Point masses of p at 0 or 1 are treated separately.

Reliability diagrams

Theorem 2.9 draws a connection from the PIT histogram to the reliability diagram or calibration curve, which is the key diagnostic tool for assessing the calibration of probability forecasts for a binary event (Murphy and Winkler, 1992; Ranjan and Gneiting, 2010).

A reliability diagram plots conditional event frequencies against binned forecast probabilities, with deviations from the diagonal indicating violations of conditional calibration.

Given a sample $\{(p_j, y_j) : j = 1, \ldots, J\}$ and a partition $0 = b_0 < b_1 < \cdots < b_I = 1$ of the unit interval into bins, we plot

$$\frac{\#\{j : p_j \in [b_{i-1}, b_i),\, y_j = 1\}}{\#\{j : p_j \in [b_{i-1}, b_i)\}} \qquad \text{vs.} \qquad \frac{1}{2}(b_{i-1} + b_i)$$

for $i = 1, \ldots, I$, and compare the resulting graph to the diagonal.

Example: Binary events

Consider a binary event Y with success probability $\Phi(a_1 + a_2)$, i.e.,

$$Y \mid a_1, a_2 \sim \text{Ber}\left(\Phi(a_1 + a_2)\right),$$

where $a_1 \sim \mathcal{N}(0, \sigma_1^2)$ and $a_2 \sim \mathcal{N}(0, \sigma_2^2)$ are independent, and Φ denotes the standard normal CDF (Ranjan and Gneiting, 2010).

Three forecasters have access to different information sets:

- forecaster 1 knows a_1 and issues $p_1 = \Phi\big(a_1/\sqrt{1+\sigma_2^2}\big)$,
- forecaster 2 knows a_2 and issues $p_2 = \Phi\big(a_2/\sqrt{1+\sigma_1^2}\big)$,
- forecaster 3 knows a_1 and a_2, and issues $p_3 = \Phi(a_1 + a_2)$.

Each forecast is ideal relative to the respective information set, and thus calibrated (Exercise).

Hint: If $X \sim \mathcal{N}(\mu, \tau^2)$, then $\mathbb{E}\,\Phi(X) = \Phi\big(\mu/\sqrt{1+\tau^2}\big)$.

Quiz

A reliability diagram lies below the diagonal for high forecast probabilities, and above the diagonal for low forecast probabilities. The probability forecasts are

A: calibrated

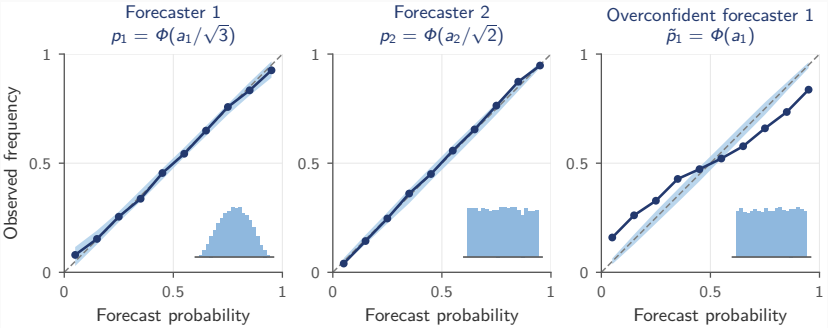
B: overconfident, i.e., too extreme

C: underconfident, i.e., not extreme enough

D: biased towards the event $Y = 1$

Example: Reliability diagrams

Reliability diagrams for a sample of size 10 000 with $\sigma_1^2 = 1$ and $\sigma_2^2 = 2$, and for an overconfident version $\tilde{p}_1 = \Phi(a_1)$ of forecaster 1, who ignores the uncertainty about a_2:



Shaded: 95% consistency bands under the hypothesis of calibration, obtained by resampling (Bröcker and Smith, 2007). Insets: histograms of the forecast values.

CORP reliability diagrams

Binning and counting can be unstable: results depend on the (ad hoc) choice of the bins, particularly for small samples or discrete forecasts.

The CORP approach (Dimitriadis et al., 2021) avoids ad hoc choices and is

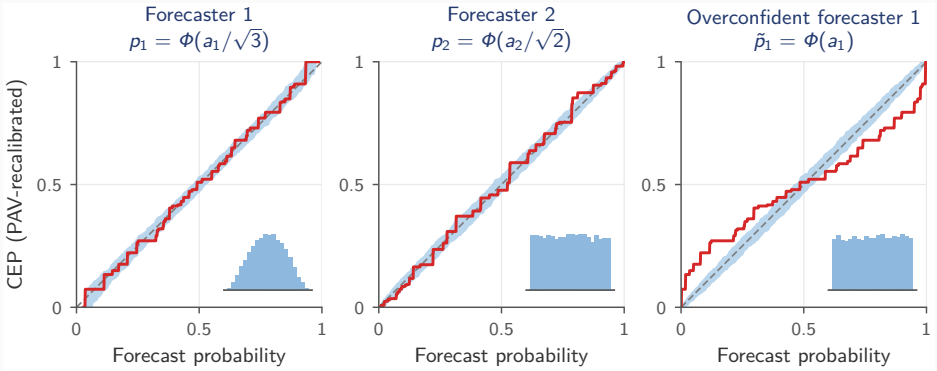
- Consistent,
- Optimally binned,
- Reproducible, and
- based on the Pool-adjacent-violators (PAV) algorithm.

The conditional event probabilities (CEPs) are estimated via isotonic regression of the binary outcomes on the forecast values, which is computed with the PAV algorithm.

Implementation: `R` package `reliabilitydiag`

Example: CORP reliability diagrams

CORP reliability diagrams for the three forecasts from before:



Shaded: 90% consistency bands under the hypothesis of calibration, obtained by resampling. Insets: histograms of the forecast values.

Auto-calibration

For non-binary outcomes, the natural generalization of conditional calibration is auto-calibration (Tsyplakov, 2011).

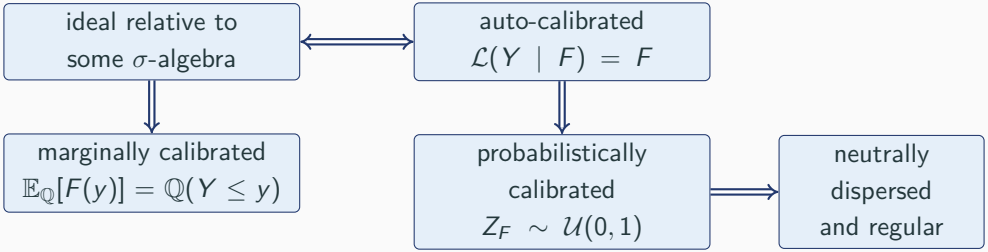
Definition 2.10: Auto-calibration

The forecast F is **auto-calibrated** if

$$\mathcal{L}(Y \mid F) = F \quad \text{almost surely.}$$

- A forecast is auto-calibrated if and only if it is ideal relative to some σ-algebra, namely, $\sigma(F)$ (Exercise).
- For a binary outcome, auto-calibration is equivalent to conditional calibration.
- For non-binary outcomes, the equivalence to probabilistic calibration fails, and auto-calibration is a much stronger requirement than probabilistic calibration.

Summary: Notions of calibration



- The converse implications fail in general, as shown by the unfocused and sign-reversed forecasts.
- For a binary outcome, probabilistic, conditional and auto-calibration are equivalent.
- The climatological forecast is calibrated in all of these senses, but not very informative $\rightarrow$ sharpness (Section 2.3)

Overview

2. Prediction Spaces, Calibration and Sharpness

2.1 Prediction spaces

2.2 Calibration

2.3 Sharpness

Calibration and sharpness

Following Murphy and Winkler (1987) and Gneiting et al. (2007), the goal in probabilistic forecasting is to

maximize the sharpness of the predictive distributions subject to calibration.

- Calibration is a multi-faceted property of the joint distribution of the forecasts and the observations, with auto-calibration being the strongest such property. Essentially, the observation is supposed to be a random draw from the predictive distribution.
- Sharpness refers to the concentration of the predictive distributions, and so is a property of the forecasts only.

  Evidently, the more concentrated, the better, subject to calibration.

Quiz

Which of the following properties of a forecasting method can be assessed without observing the outcomes?

A: marginal calibration

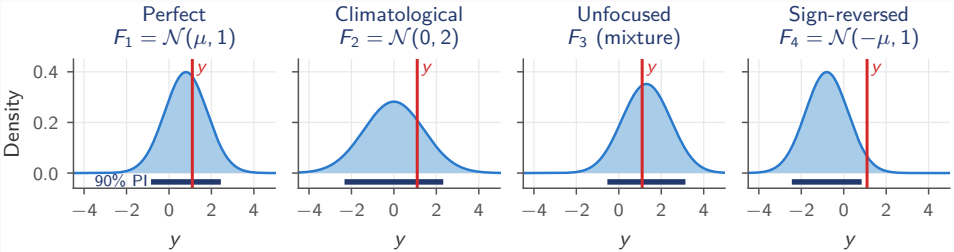
B: probabilistic calibration

C: dispersion, i.e., $\text{Var}(Z_F)$

D: sharpness

Example: Density forecasts (continued)

Predictive densities for a single forecast case ($\mu = 0.8$, $\tau = 1$), with the observation y (red) and the central 90% prediction interval (PI, dark blue):



The perfect, climatological and unfocused forecasts are all probabilistically calibrated, but the perfect forecast is the sharpest.

The sign-reversed forecast is equally sharp, but not calibrated.

Sharpness of probability forecasts

In the case of probability forecasts of a binary event, a sharp forecast tends to be close to 0 or 1.

Recall the binary example from Section 2.2.4 with $\sigma_1^2 = 1$ and $\sigma_2^2 = 2$, where

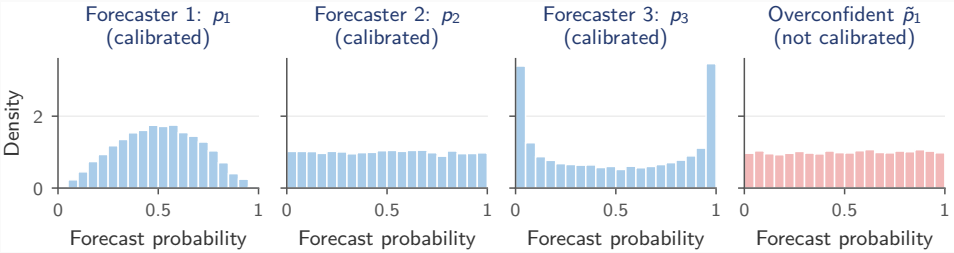
- forecaster 1 knows a_1 and issues $p_1 = \Phi\big(a_1/\sqrt{1+\sigma_2^2}\big)$,
- forecaster 2 knows a_2 and issues $p_2 = \Phi\big(a_2/\sqrt{1+\sigma_1^2}\big)$,
- forecaster 3 knows a_1 and a_2, and issues $p_3 = \Phi(a_1 + a_2)$,

and $\tilde{p}_1 = \Phi(a_1)$ is an overconfident version of forecaster 1.

Forecasts p_1, p_2 and p_3 are ideal, and thus calibrated.

Example: Sharpness of probability forecasts

Histograms of the forecast probabilities for a sample of size 10 000:



All three ideal forecasts are calibrated, but p_3 is sharper than p_2, which is sharper than p_1: the more information, the sharper the ideal forecast.

The overconfident forecast $\tilde{p}_1$ is as sharp as p_2, but not calibrated (cf. its reliability diagram in Section 2.2.4).

Sharpness of density forecasts

In the case of density forecasts of a real-valued variable, we can assess sharpness in terms of the width of prediction intervals.

For example, the central 90% prediction interval ought to be as short as possible, subject to its empirical coverage being at the nominal 90% level, with 5% of the observations lying to its left and right, respectively.

Competing forecasts can be compared by the average width of their prediction intervals or, if the widths vary between forecast cases, by boxplots of the widths (sharpness diagram).

For a continuous predictive CDF F, the observation falls into the central $(1 - \alpha)$ prediction interval if and only if $Z_F \in [\frac{\alpha}{2}, 1 - \frac{\alpha}{2}]$. Hence, probabilistically calibrated forecasts attain the nominal coverage.

Example: Density forecasts (continued)

Width and coverage of the central 50% and 90% prediction intervals:

	50% interval		90% interval	
Forecast	Width	Coverage	Width	Coverage
Perfect	1.35	50.0%	3.29	90.0%
Climatological	1.91	50.0%	4.65	90.0%
Unfocused	1.52	50.0%	3.68	90.0%
Sign-reversed	1.35	23.7%	3.29	53.8%

Among the calibrated forecasts, the perfect forecast is the sharpest. The sign-reversed forecast is equally sharp, but its prediction intervals are far from the nominal coverage.

Example: 2-step ahead forecasts of an AR(1) process (continued)

We return to the AR(1) process $Y_t = \varphi Y_{t-1} + \epsilon_t$ with $\varphi = 0.8$ and iid $\epsilon_t \sim \mathcal{N}(0, 1)$ from Section 2.2.3, and compare four forecasts at lead time $k = 2$:

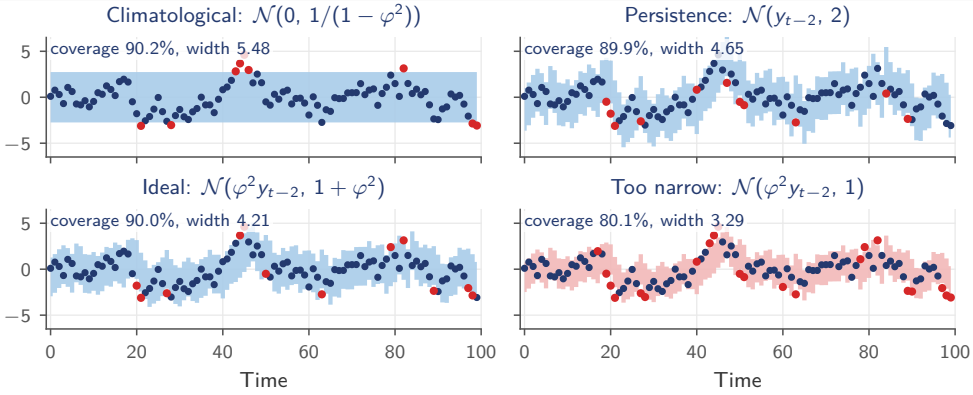
- climatological: $\mathcal{N}\left(0, 1/(1-\varphi^2)\right)$
- persistence (probabilistic no-change): $\mathcal{N}(y_{t-2}, 2)$
- ideal: $\mathcal{N}(\varphi^2 y_{t-2}, 1+\varphi^2)$
- too narrow: $\mathcal{N}(\varphi^2 y_{t-2}, 1)$, which ignores the uncertainty due to ϵ_{t-1}

The climatological, persistence and ideal forecasts are probabilistically calibrated.

For the persistence forecast, use that $Y_t - Y_{t-2} \sim \mathcal{N}(0, 2)$ (Exercise).

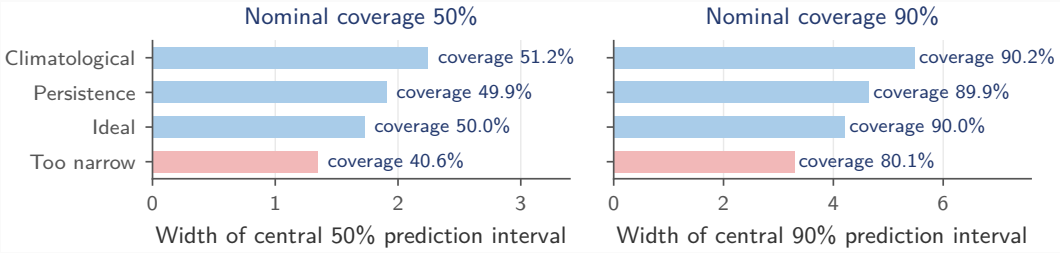
Example: 2-step ahead forecasts of an AR(1) process (continued)

Central 90% prediction intervals for 100 time steps (observations outside the interval in red); coverage and width are computed from 5 000 forecast cases:



Example: 2-step ahead forecasts of an AR(1) process (continued)

Width and empirical coverage of the central 50% and 90% prediction intervals:



The climatological, persistence and ideal forecasts are (empirically) calibrated, and the ideal forecast is the sharpest.

The "too narrow" forecast is even sharper, but its prediction intervals fall short of the nominal coverage.

Quiz

Forecast A issues central 90% prediction intervals with an average width of 3.3 and an empirical coverage of 80%. For forecast B, the average width is 4.2 and the empirical coverage is 90%. Which statement is correct?

A: Forecast A is preferable, since it is sharper.

B: Forecast B is preferable, since forecast A is not calibrated.

C: Both forecasts are equally good.

D: Neither, since the climatological forecast is always preferable.

Calibration and sharpness: Summary

calibrated and sharp perfect forecast, ideal AR forecast, forecaster 3 in the binary example	**calibrated, but not sharp** climatological forecasts
sharp, but not calibrated sign-reversed forecast, overconfident forecasts	**neither** e.g., overdispersed forecasts

To address calibration and sharpness simultaneously, we use proper scoring rules (Gneiting and Raftery, 2007), to be discussed in Section 3.

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